



KAZAKH-BRITISH
TECHNICAL
UNIVERSITY

Kazakh-British Technical University
School of Information Technology and Engineering

DISTRIBUTED CONTROL FRAMEWORK OF MULTIROBOT SYSTEM USING AUTOMATION AND CONTROL APPLICATIONS

Authors:

Abdimalik A., Zeinulla R.,
Rakhmetkali A., Masimba C.
Timur Samigulin

Almaty, 2026

CONTENT

- 1 | Introduction
- 2 | Technological process
- 3 | Control Object
- 4 | Mathematical model of the control object
- 5 | Control object regulation
- 6 | Complex of equipment and technical implementation
- 7 | Program realization
- 8 | Economic part
- 9 | Labor protection
- 10 | Designed drawings (AutoCAD)
- 11 | Certificate and other achievements

INTRODUCTION AND STATEMENT OF PROBLEM

According to OSHA, paint vapors contain volatile organic compounds (VOCs) that can cause breathing problems and skin irritation during long-term exposure. Manual painting is also inefficient and leads to unstable coating quality. To improve safety, accuracy, and repeatability, an automated multi-robot painting system with real-time monitoring is required. The novelty of this work is a PLC–MATLAB-based control architecture, where MATLAB handles high-level coordination and optimization, while an industrial PLC ensures real-time control and synchronization of multiple robot manipulators.

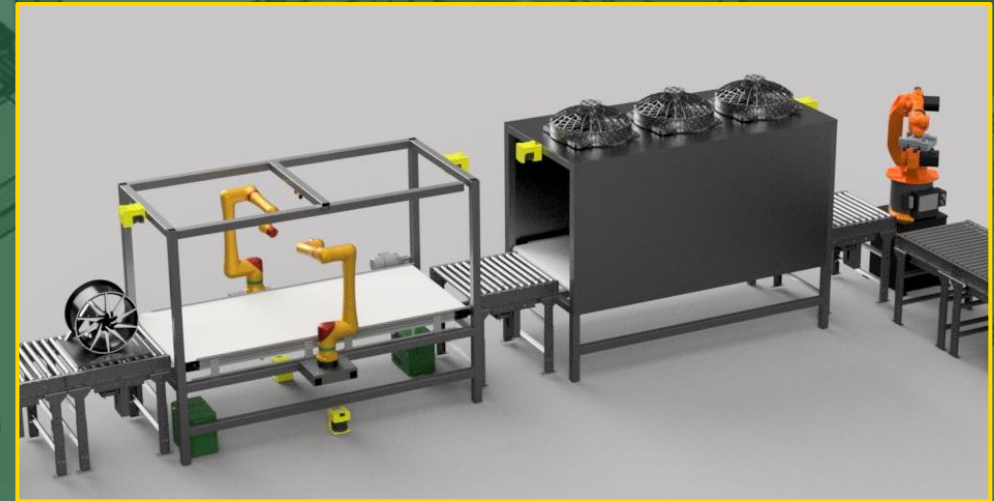


Figure 1 – Industrial Scale Process

Scale of the technological Process

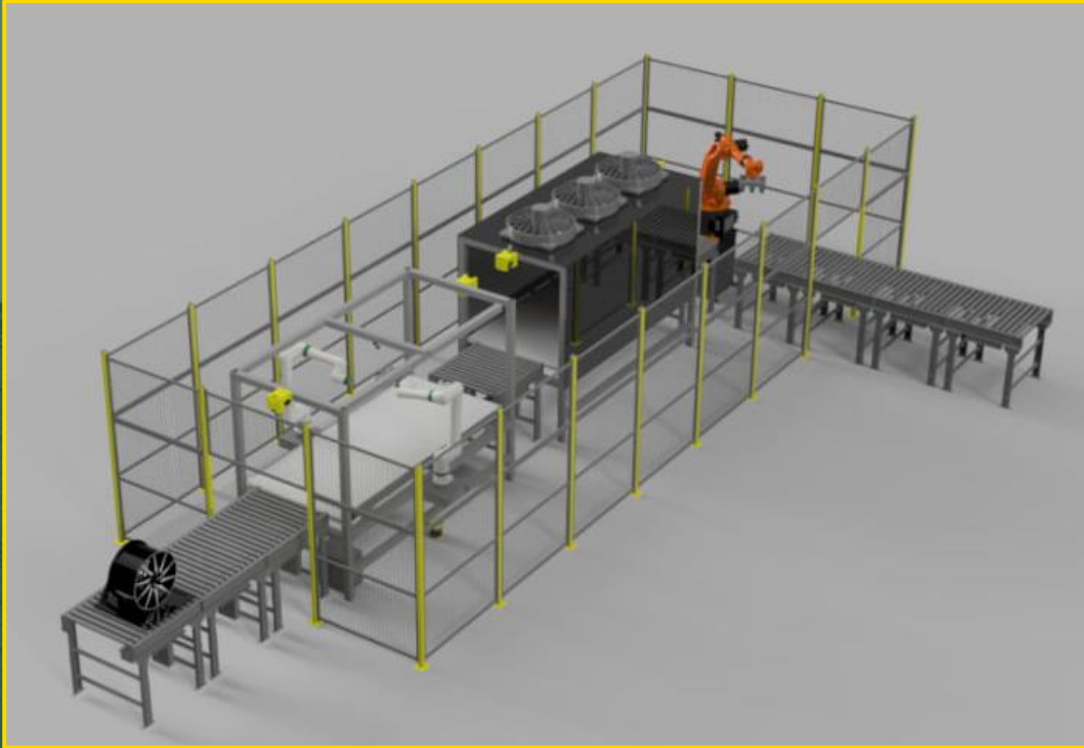


Figure 2 – Industrial Scale Process



Figure 3 – Laboratory Scale Process

Figure 2 presents the **industrial-scale implementation** of the painting process, while Figure 3 shows the **laboratory-scale prototype** developed for testing and validation. The prototype reproduces the key technological operations and control functions of the industrial system.

Description of the technological process

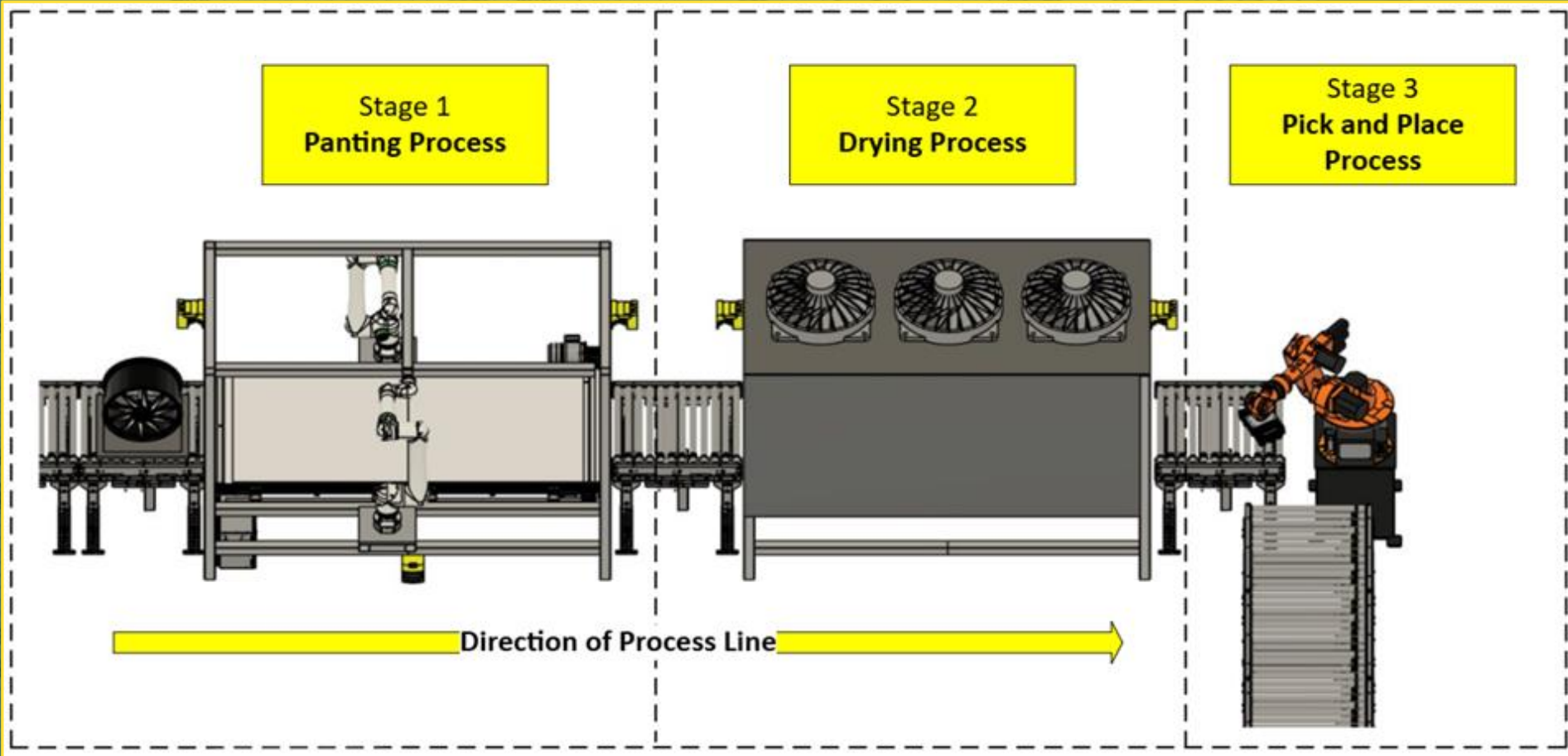


Figure 4 – The Operation workflow of the Industrial Technological Process

TECHNOLOGICAL PROCESS

Figure 5 shows the block diagram of the automated multi-robot painting system.

The system **sets parameters** and **checks status**, then places an object on the conveyor. The conveyor moves and stops at each station: **painting** (two RoArm robots paint), **drying** (fan arch dries and verifies) and final **handling**. At the end, a KUKA robot **picks and places** the finished part, **records the result**, and the control system coordinates all movements to **keep timing stable**.

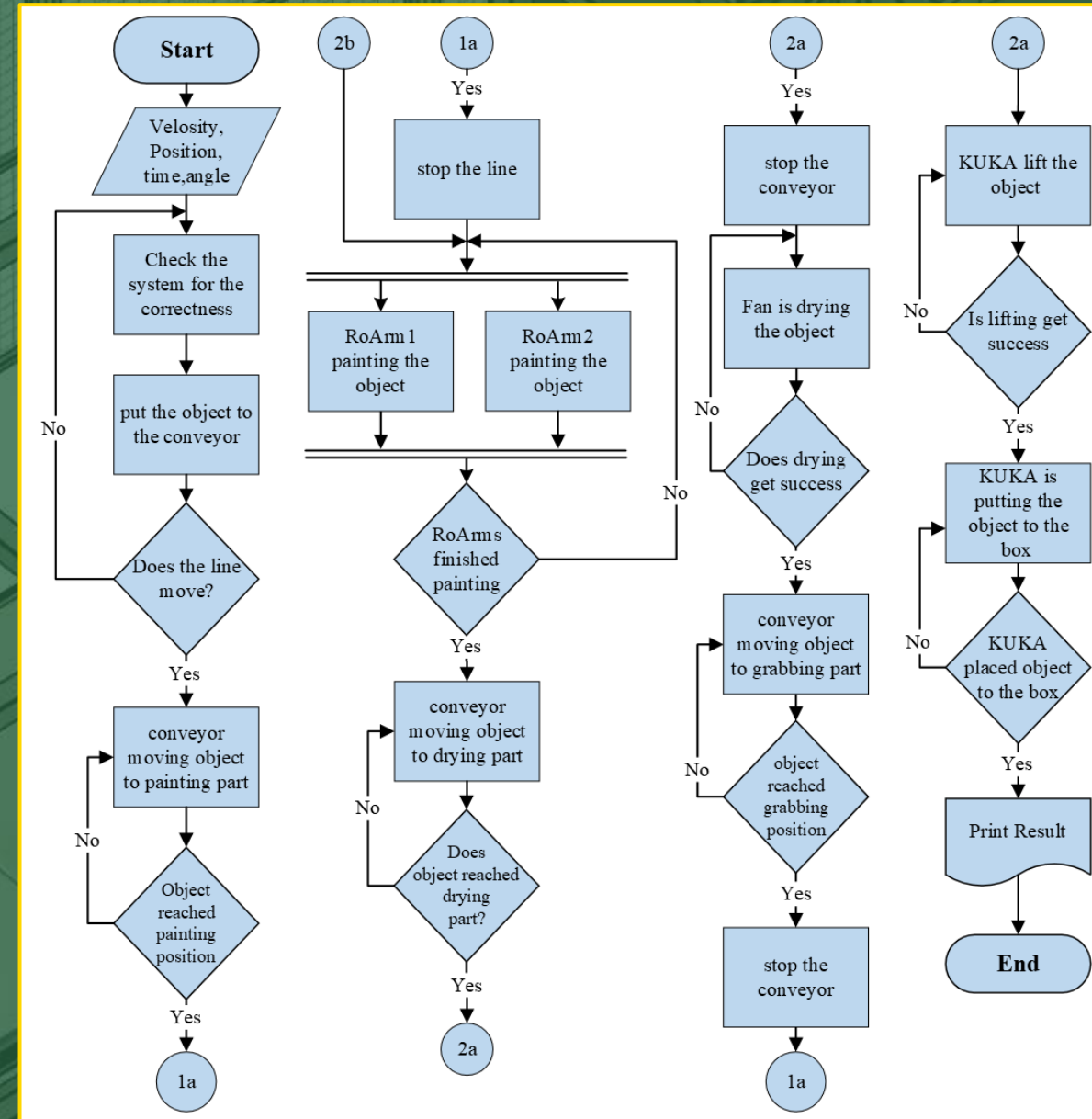
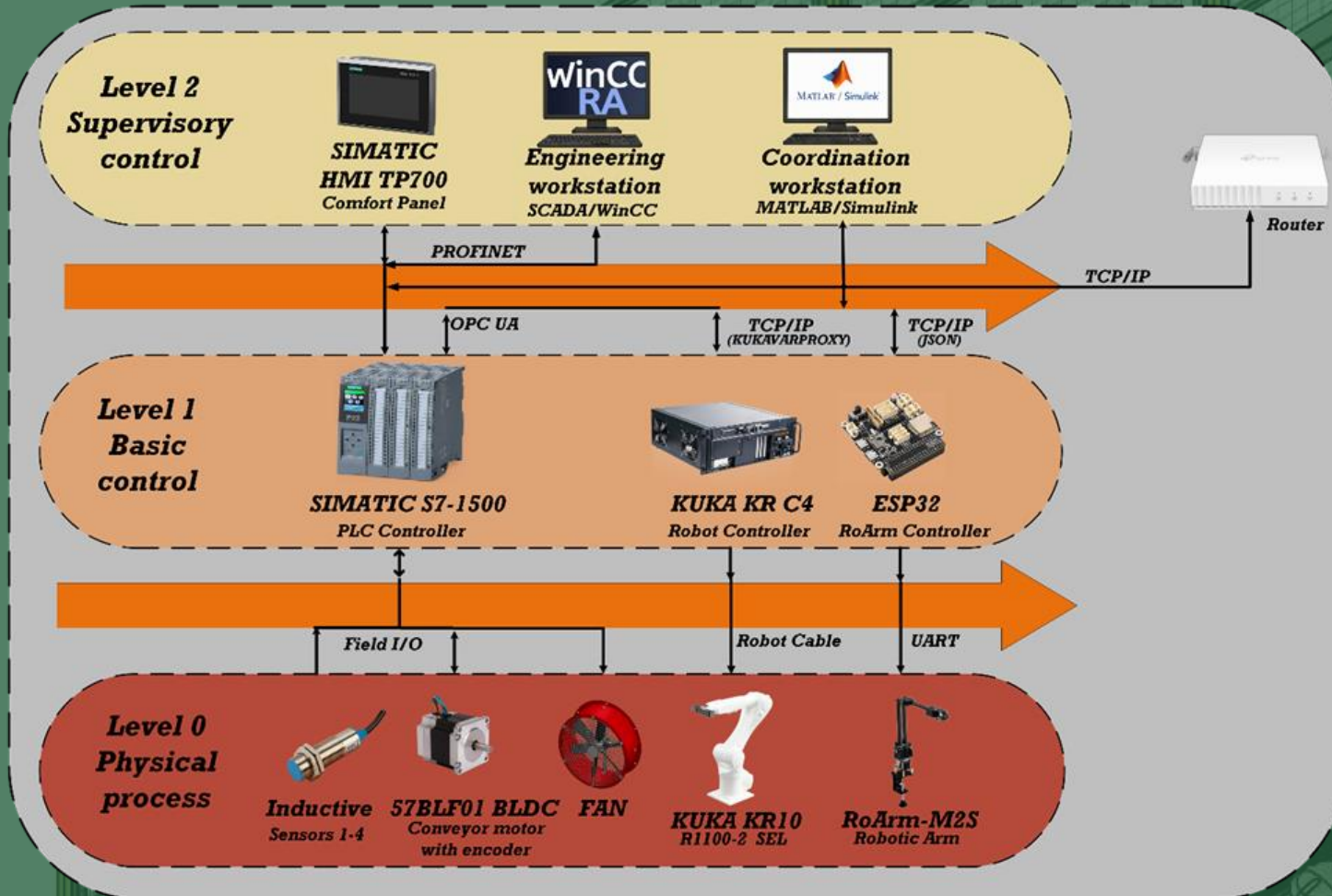


Figure 5 – Block Diagram

COMPLEX OF THE TECHNICAL DEVICES



This figure illustrates the **three-level control and communication architecture** of the automated painting system. **Level 0** represents the **physical process**, including sensors, the BLDC conveyor motor, fan and robotic manipulators. **Level 1** provides **basic control** through the SIMATIC S7-1500 PLC, KUKA robot controller and ESP32-based RoArm controller. **Level 2** represents **supervisory control**, where the HMI panel, SCADA/WinCC workstation and MATLAB/Simulink coordination workstation are used for monitoring, visualization and high-level coordination. **Communication between the devices** is organized through **PROFINET, OPC UA, TCP/IP, UART** and field I/O connections. Inputs:5; Outputs:2.

Figure 6 - Purdue model-based architecture of the automated painting system

COMPLEX OF EQUIPMENT: SIEMENS S7-1500 PLC

Parameter	SIMATIC S7 – 1500	Modicon M340
Number of Points	1024	1024
Memory	1.5 Mb for data	4096
Memory Card	Yes	No
Programming Language	LAD, FBD, SCL, STL, S7-Graph	LD, FBD, SFC, ST, IL
Programming Tools	TIA Portal	Unity Pro / EcoStruxure Control Expert
Network	PROFINET, Modbus TCP, OPC UA, Web Server	Modbus TCP, CANopen
OPC UA	Supported	Requires Additional OPC UA communication module
Manufacturer	Siemens	Schneider Electric

Table 1 – Comparison of Controllers



Figure 7 – S7-1513F-1 with I/O modules

COMPLEX OF EQUIPMENT: DIGITAL INPUT MODULE

Parameter	Characteristics
Product Type	Digital Input module
Input Type	Current sink
Discrete Input Number	32
Discrete Input Voltage	24V DC Positive
Discrete Input Current	2.5 mA
Operating Temperature	0°C ... +60°C
Storage Temperature	−40°C ... +70°C
IP Degree of Protection	IP20
Mass	0.260 kg
Dimensions	35 mm × 147 mm × 129 mm

Table 2 – Characteristics of digital input module



Figure 8 – Digital input module

COMPLEX OF EQUIPMENT: DIGITAL OUTPUT MODULE

Parameter	Characteristics
Product Type	Digital Output module
Output Type	Transistor
Discrete Output Number	32
Discrete Output Voltage	24V DC Positive
Discrete Output Current	0.5 mA
Operating Temperature	0°C ... +60°C
Storage Temperature	−40°C ... +70°C
IP Degree of Protection	IP20
Certificates	PL d, SIL 2
Mass	0.260 kg
Dimensions	35 mm × 147 mm × 129 mm

Table 3 – Characteristics of digital output module



Figure 9 – Digital output module

COMPLEX OF EQUIPMENT: SIMATIC HMI TP700

Specification	Value
Model	6AV2124-0GC01-0AX0
Type	HMI Comfort Panel
Display	7-inch widescreen TFT
Resolution	800 × 480 pixels
Colors	16 million
Configuration memory	12 MB
Recipe memory	2 MB
Operating system	Windows CE 6.0
Supply voltage	24 V DC
Current consumption	0.5 A
Power consumption	12 W
Communication interfaces	1 × PROFINET (2-port switch), 1 × MPI/PROFIBUS DP
Additional interfaces	1 × RS 422/485, 2 × USB 2.0, 2 × SD card slots

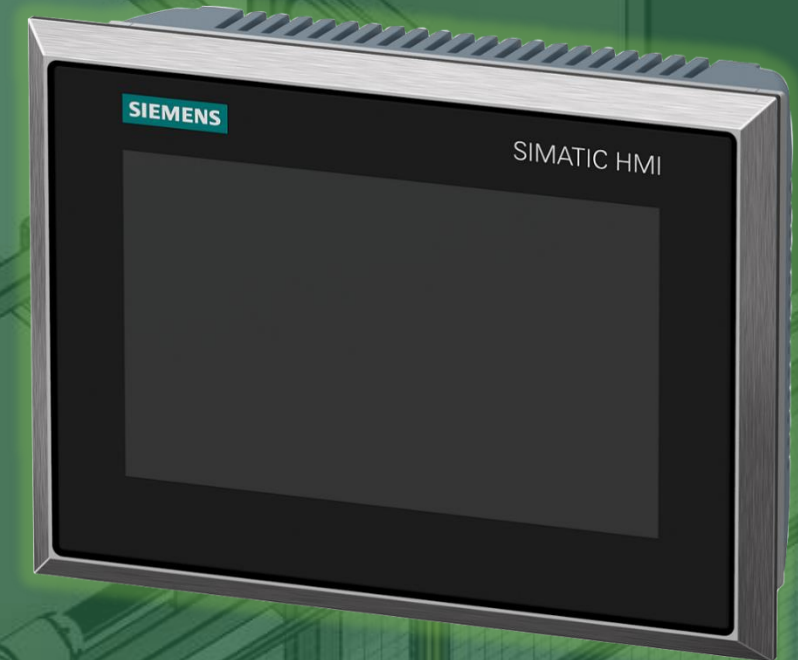


Figure 10 – SIMATIC HMI TP700 Comfort Panel

Table 4 – Characteristics of SIMATIC HMI TP700 Comfort Panel

COMPLEX OF EQUIPMENT: INDUCTIVE SENSOR

Specification	Value
Model	3RG40 22-3JB00
Type	Inductive proximity sensor (2-wire, PLC type)
Operating voltage (DC)	15 ... 34 V DC
Rated operating distance (Sn)	4 mm (not flush)
Output type	2-wire, NO (normally open)
Protection degree	IP67
Connection type	M12 connector / 2 × 0.25 mm ² cable
Housing dimensions	M12 × 40 mm
Mounting distance	≥ 15 mm between sensors
Power-up delay (tv)	40 ms



Figure 11 – SIMATIC PXI200 M12 NOT
flush

Table 5 – Characteristics of Inductive proximity sensor

COMPLEX OF EQUIPMENT: RoArm-M2-S

Specification	Value
1	2
Power Input	12V, 5A power supply
Weight	973.5±15g
Payload	0.2 kg at 0.5 m
Work Space (Horizontal)	diameter: 1090mm, 360° omnidirectional
Work Space (Vertical)	798mm
Accuracy	± 5 mm (at the same load)
Degree of freedom	4
Controller	ESP32-WROOM-32
Range of motion of Base	360°
Range of motion of Shoulder	180°
Range of motion of Elbow	225°
Range of motion of Hand	135°/270°
Servo Rotation Speed	40rpm (no torque limit)

Table 6 - Technical characteristics of
RoArm-M2-S



Figure 12 – RoArm-M2-S

COMPLEX OF EQUIPMENT: KUKA KR10 R1100-2/SEL

Specification	Value
1	2
Power Input	220-240 V
Weight	59 kg
Payload	10 kg
Maximum Payload	11 kg
Maximum range	1101 mm
Accuracy	±0,02 mm
Degree of freedom	6
Temperature range	0°C-45°C
Controller	C4SR
Work Space	208×208 mm
Range of motion along the axes 1	±170°
Range of motion along the axes 2	-190°/45°
Range of motion along the axes 3	-120°/156°
Range of motion along the axes 4	±185°
Range of motion along the axes 5	±120°
Range of motion along the axes 6	±350°

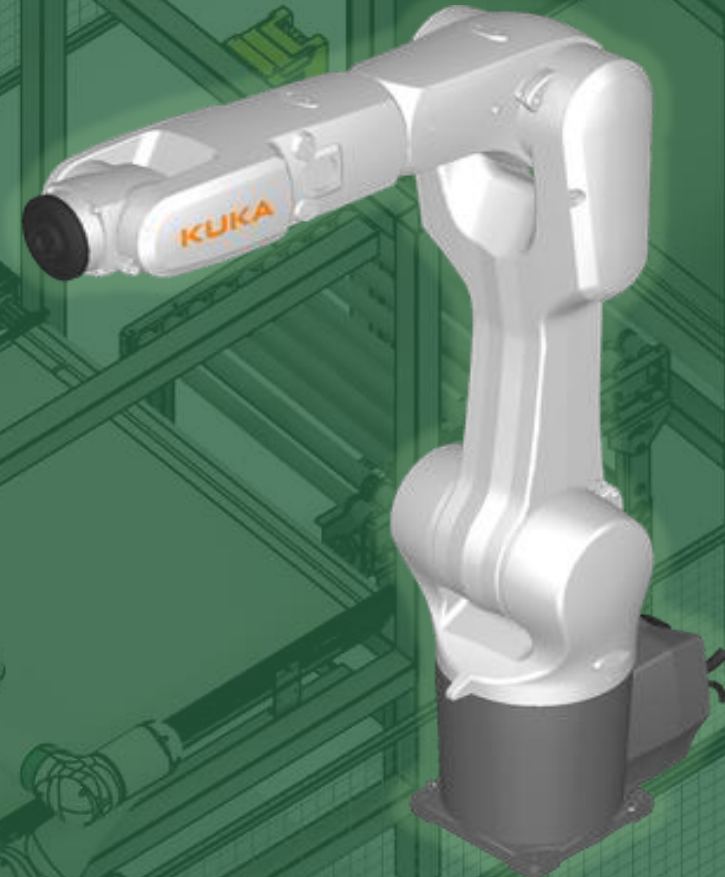


Figure 13 – KUKA KR10 R1100-2/SEL

Table 7 - Technical characteristics of KUKA KR10 R1100-2 SEL

COMPLEX OF EQUIPMENT: BLDC motor

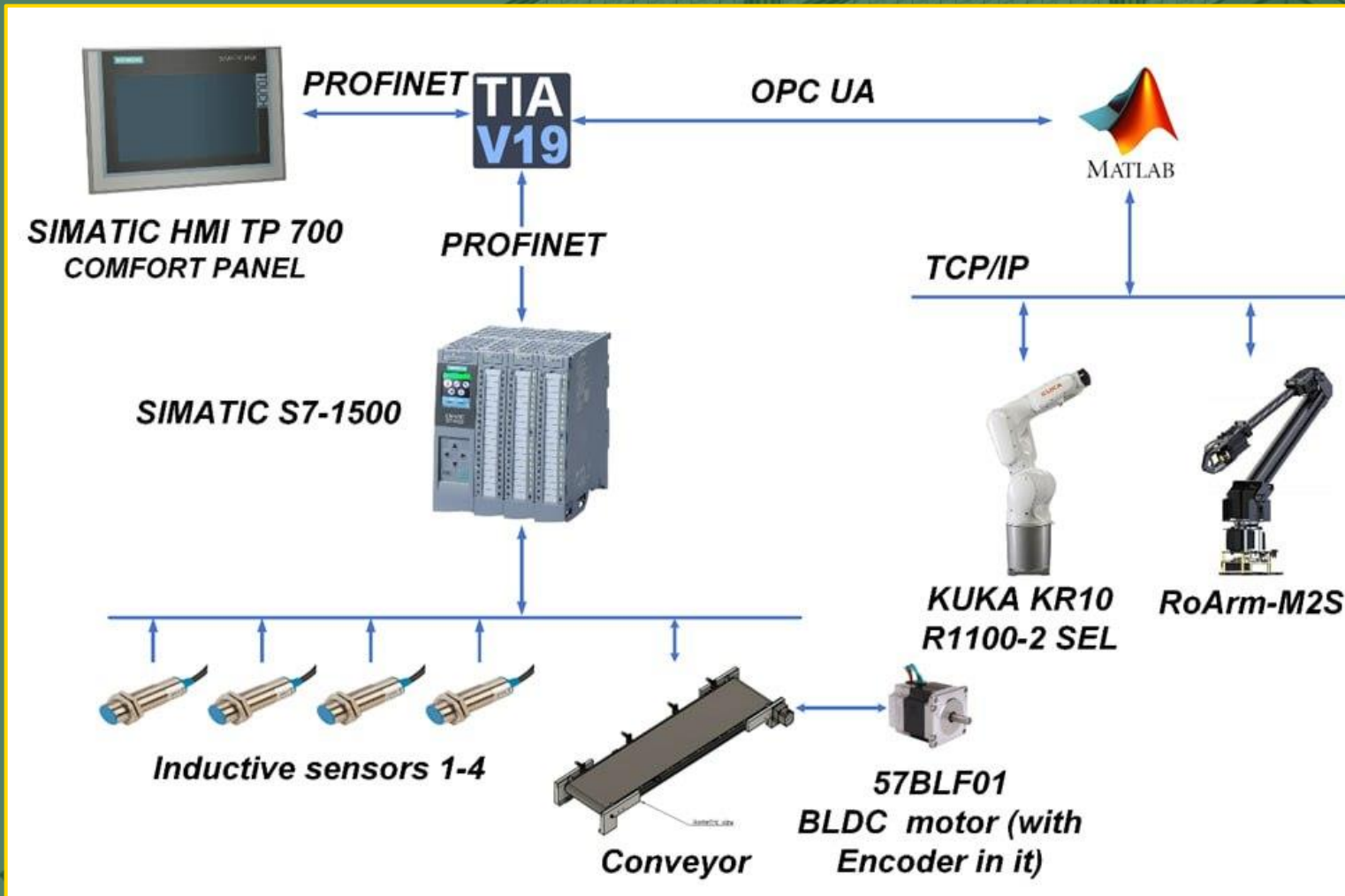
Specification	Value
1	2
Rated Voltage	24 V DC
Rated Current	4 A
Rated Speed	3000 RPM
Number of Poles	8
Number of Phases	3
Output Power	63 W
Rated Torque	0.2 Nm
Winding Type	Star
Mass	0.65 kg
Body Length	59 mm
Ambient Temperature Range	-20°C to 50°C

Table 8 - Technical characteristics of
BLDC motor



Figure 14 – 57BLF01 BLDC motor

COMPLEX OF EQUIPMENT AND TECHNICAL IMPLEMENTATION



- Siemens S7-1500 PLC for process control
- MATLAB/Simulink for robot coordination
- Two RoArm-M2-S manipulators for painting
- KUKA KR10 for final pick-and-place
- Conveyor system synchronization
- HMI for state of industrial process

Figure 15 – Architectural scheme

PROGRAM REALIZATION OF MANIPULATORS

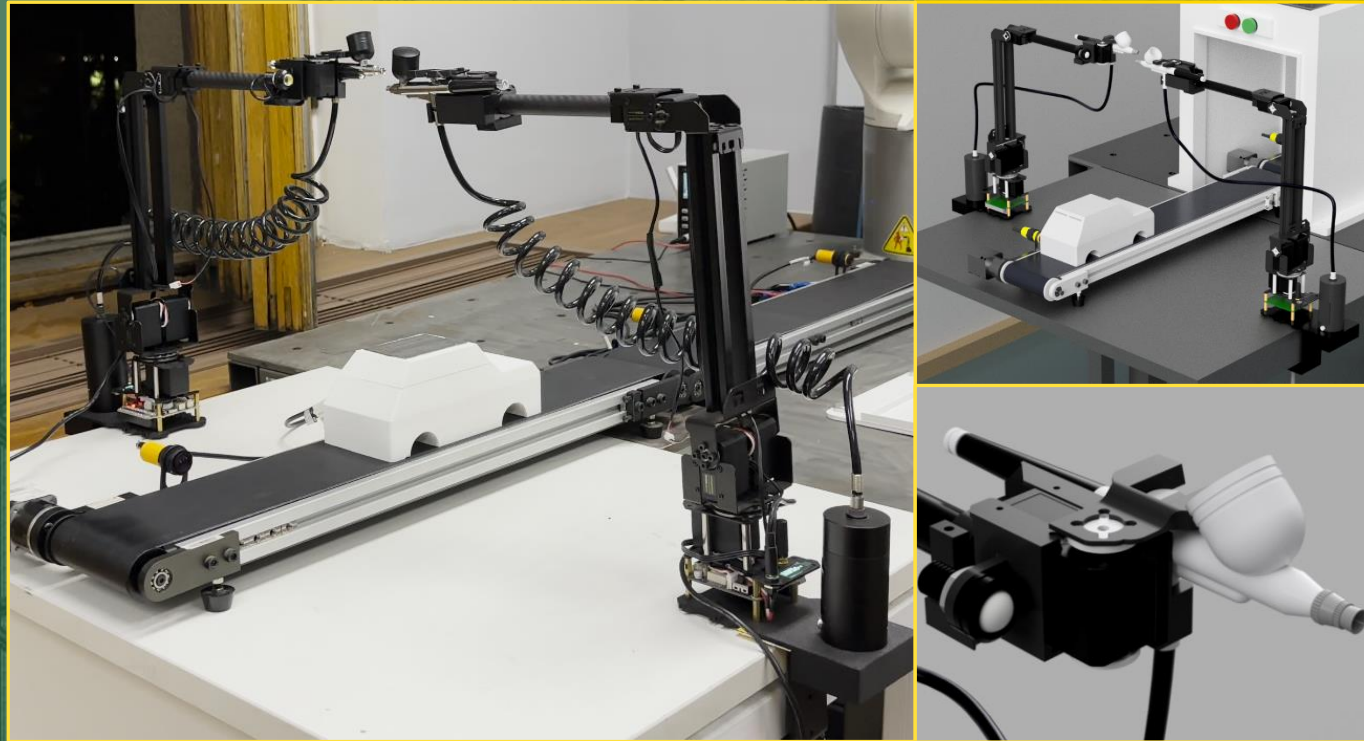


Figure 16 – Painting process

Statement of the problem painting process to construct and program cooperated painting system with two RoArm M2-S manipulators with MATLAB software. For precise painting of the car details, to show real time telemetry.

- End effector 3D modeled in Fusion360
- Custom spraying end effector
- Programming in MATLAB
- Trajectory generation

PROGRAM REALIZATION OF MANIPULATORS

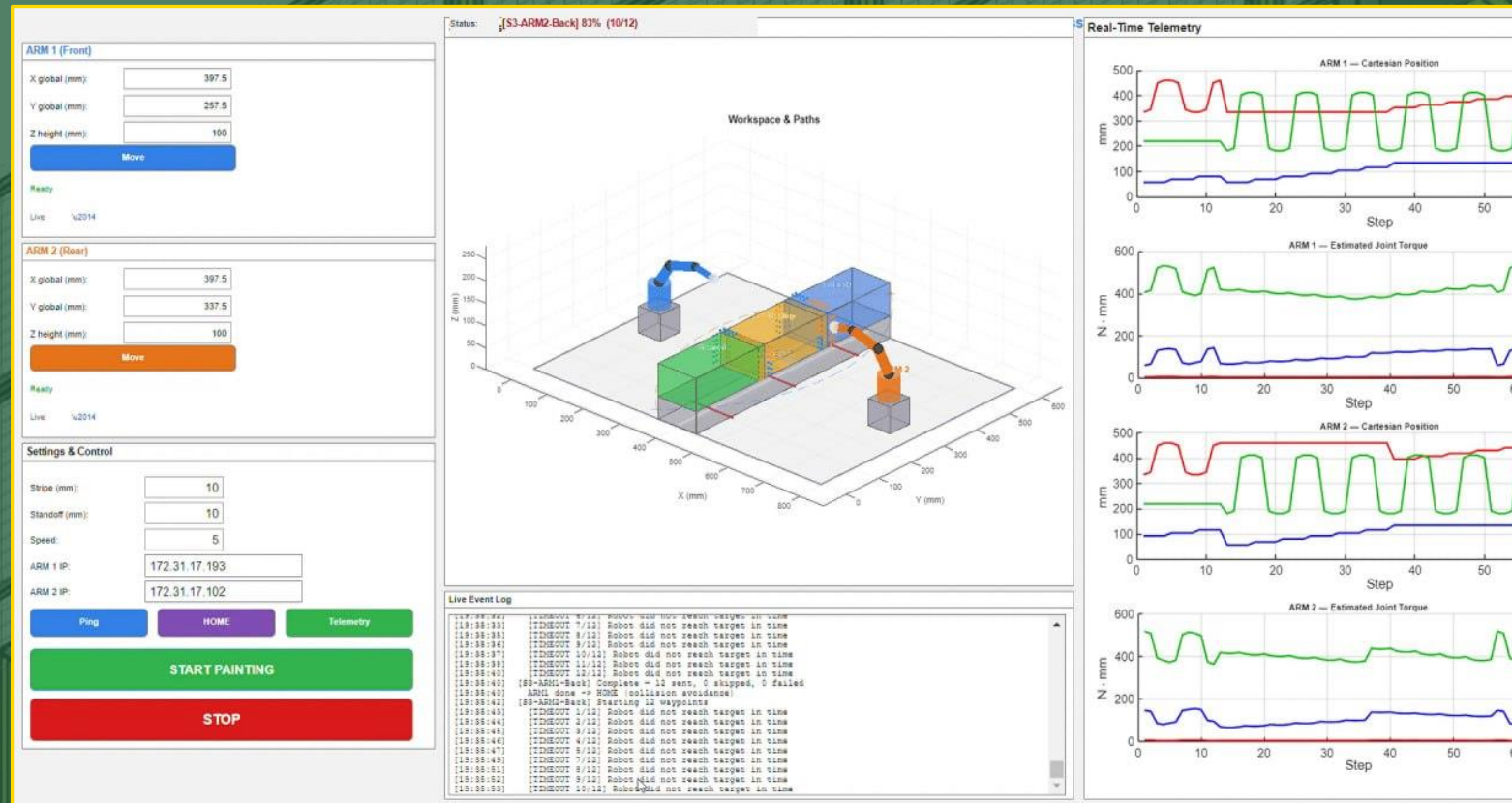


Figure 17 – Painting process GUI

Two RoArm-M2-S manipulators painting process
Wi-Fi HTTP/JSON communication with MATLAB
3D preview of workspace and painting path
Synchronized movement of two manipulators
Real-time telemetry: Cartesian position and estimated joint torque
Start, stop, home position and telemetry control

PROGRAM REALIZATION OF MANIPULATORS

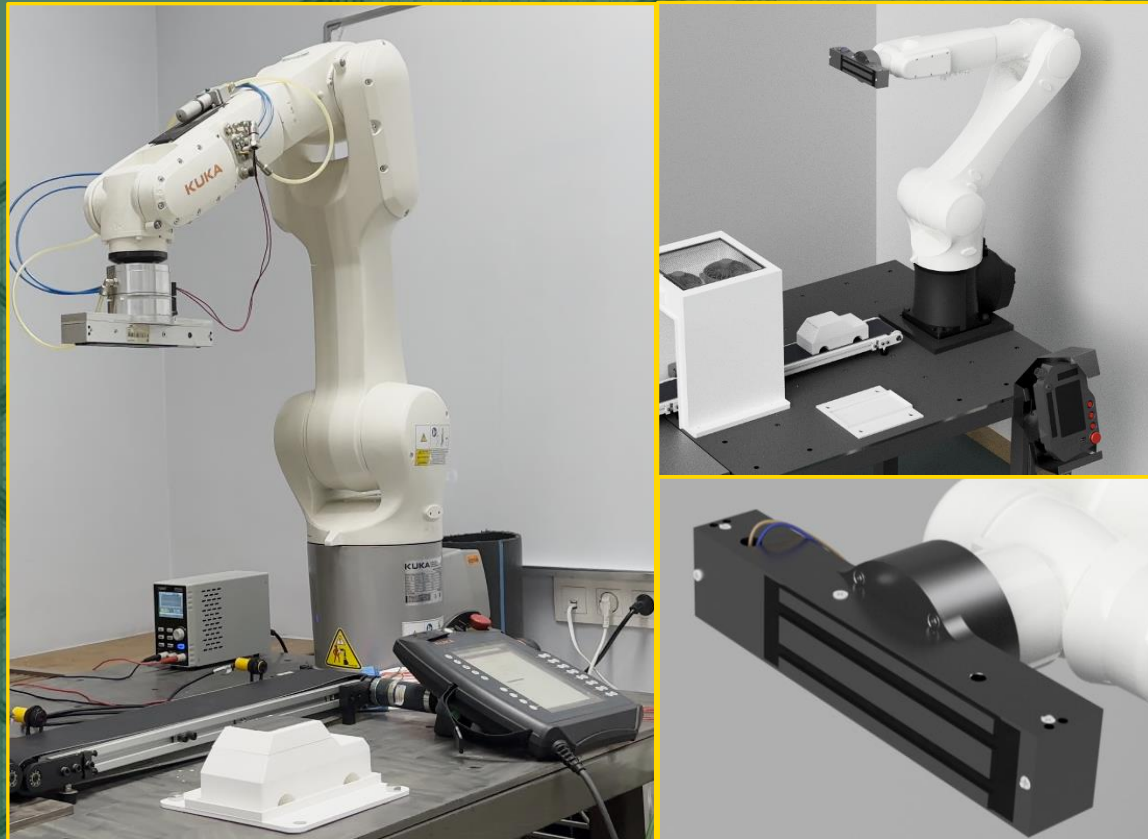


Figure 18 – Pick and Place process

Statement of the problem pick and place process to construct and program with industrial KUKA KR10 R1100-2 manipulator with MATLAB software. For smooth transportation process, between pick and place point positions.

- End effector 3D modeled in Fusion360
- Custom magnetic end effector
- Programming in MATLAB
- Pick and Place Trajectory generation

PROGRAM REALIZATION OF MANIPULATORS

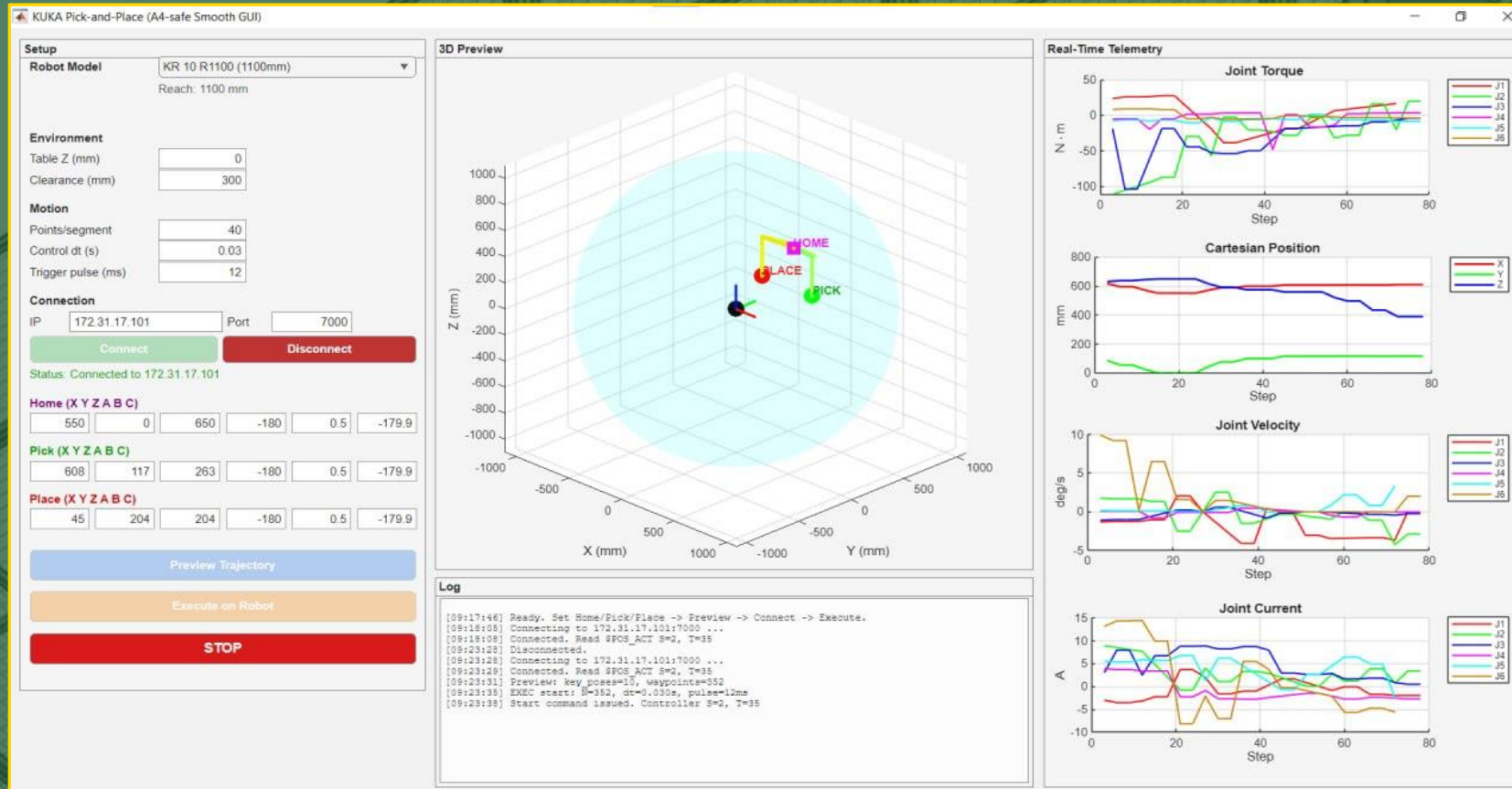


Figure 19 – Pick and Place process GUI

KUKA KR10 R1100-2 manipulator pick-and-place process

TCP/IP communication with MATLAB through KukaVarProxy

3D preview of pick, approach, lift and place trajectory

Real-time telemetry: joint torque, Cartesian position, joint velocity and joint current

Emergency stop and execution status monitoring

PROGRAM REALIZATION OF MANIPULATORS

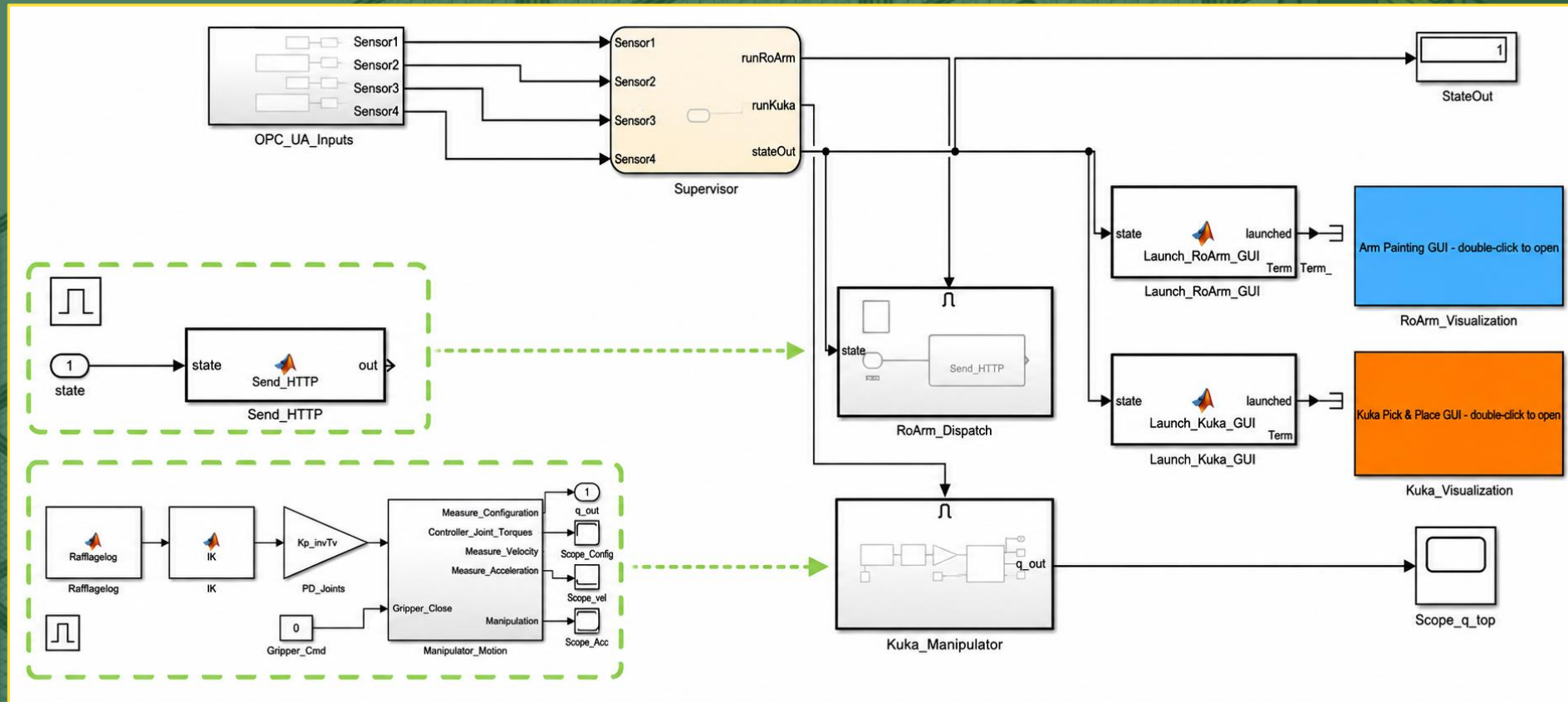


Figure 20 – Simulink Program realization

Simulink is used as the **supervisory logic layer** of the robotic process. It receives **sensor** and **process signals OPC UA**, checks the current system state, and **launches** the required **robot subsystem**. The **RoArm** block starts the **painting GUI** and painting motion. The **KUKA** block starts the **pick-and-place GUI** and final handling process. This structure allows the complete automated cycle to be started from one control command.

PROGRAM REALIZATION OF MANIPULATORS

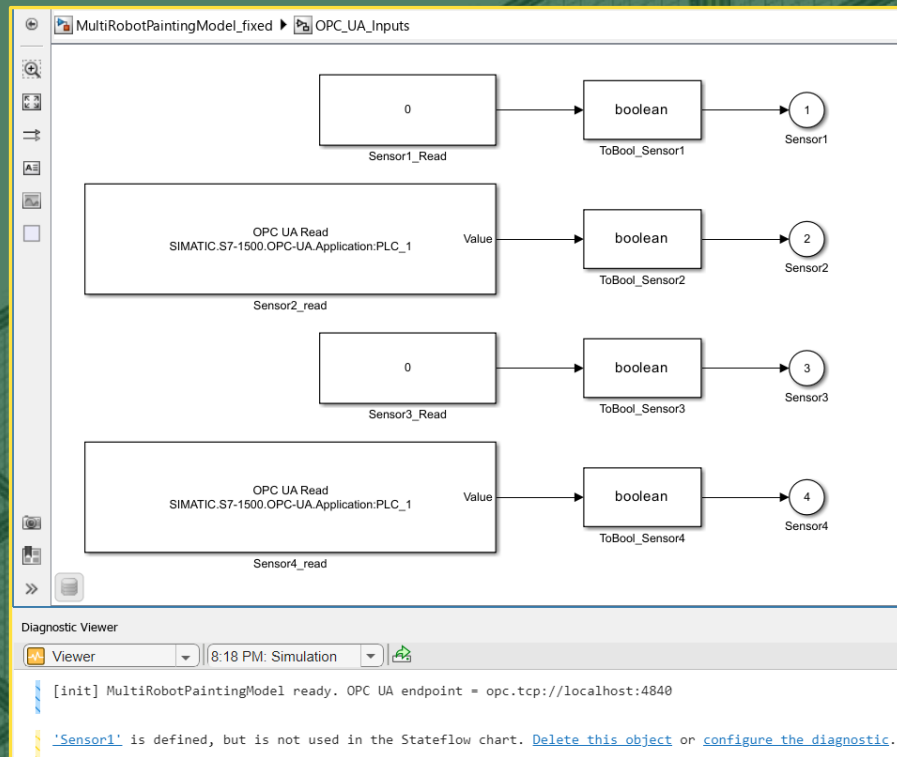


Figure 21 – Simulink OPC UA input signal reading

PLC_stage_DIPLOMA > PLC_1 [CPU 1513F-1 PN] > Watch and force tables > Watch table_1					
	i	Name	Address	Display format	Monitor value
1		"Send_value1_fr...	%M11.0	Bool	☒ TRUE
2		"Send_value2_fr...	%M11.1	Bool	☐ FALSE
3		"Read_value1_to...	%M11.2	Bool	☒ TRUE
4		"Read_value2_to...	%M11.4	Bool	☐ FALSE

OPC UA server interface			
	Browse name		Access level
1	▼ OPC-UA		Interface
2	■	Read_value1_to_Matlab	RD/WR
3	■	Read_value2_to_Matlab	RD/WR
4	■	Send_value1_from_Matlab	RD/WR
5	■	Send_value2_from_Matlab	RD/WR

Figure 22 — PLC tags configured for OPC UA communication

The PLC tags are configured through the **OPC UA** interface to exchange **Boolean signals** with **MATLAB/Simulink**. **Read_value** signals send the current PLC state to **Simulink**, while **Send_value** signals return control commands back to the **PLC**. This provides a simple communication layer for supervising the manipulator program.

MATHEMATICAL MODEL OF THE KUKA

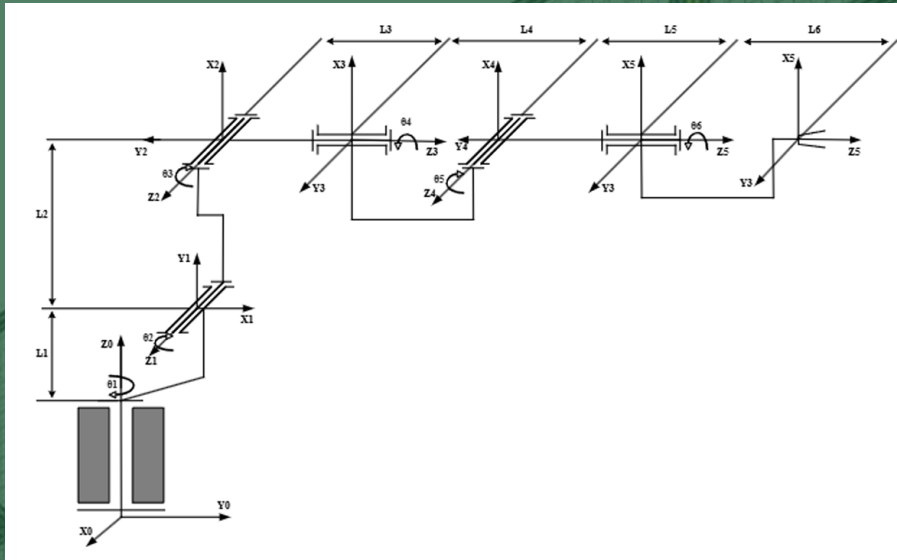


Figure 23 – Kuka kinematics model

Forward kinematic chain

$$f(q) = {}^0T_6(q) = {}^0T_1(q_1) \cdot {}^1T_2(q_2) \cdot {}^2T_3(q_3) \cdot {}^3T_4(q_4) \cdot {}^4T_5(q_5) \cdot {}^5T_6(q_6)$$

Nonlinear least-squares estimation

$$q^* = \arg \min_{(q)} \frac{1}{2} e(q)^T W e(q)$$

The **DH parameters** define the geometry of each **KUKA manipulator** link. The **forward kinematic chain** calculates the end-effector pose from the joint angles, while the **inverse kinematics objective** finds the joint configuration that minimizes the pose error for the required position and orientation.

DH transformation matrix

$$T_i^{i-1}(q_i) = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cdot \cos\alpha_i & \sin\theta_i \cdot \sin\alpha_i & a_i \cdot \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cdot \cos\alpha_i & -\cos\theta_i \cdot \sin\alpha_i & a_i \cdot \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Table 9 – Denavit–Hartenberg parameters

i	a_i [mm]	α_i [rad]	d_i [mm]	$\theta_{0,i}$ [rad]
1	350	$-\pi/2$	675	0
2	850	0	0	0
3	0	$\pi/2$	0	0
4	0	$-\pi/2$	-820	0
5	0	$\pi/2$	0	0
6	0	π	-170	0

PROGRAM REALIZATION OF SCADA

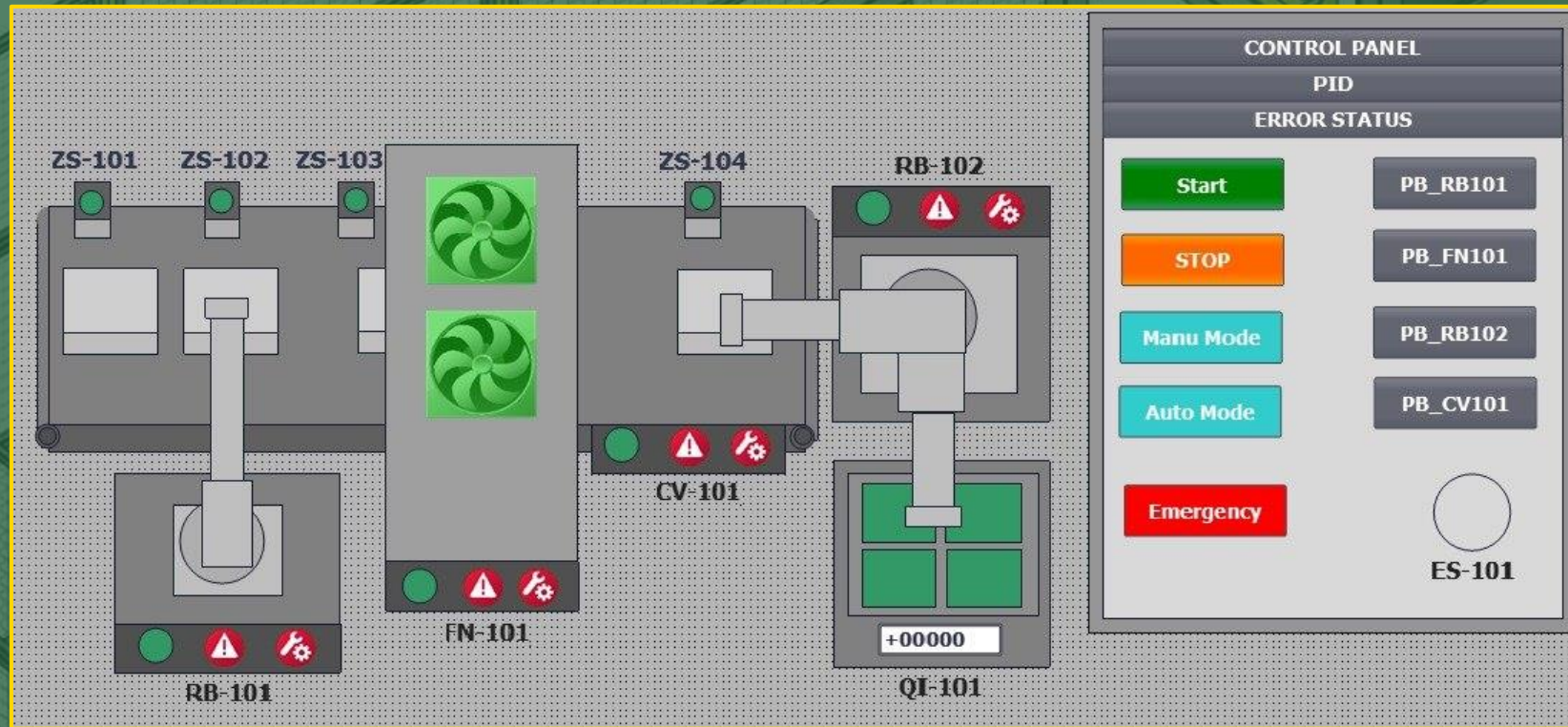
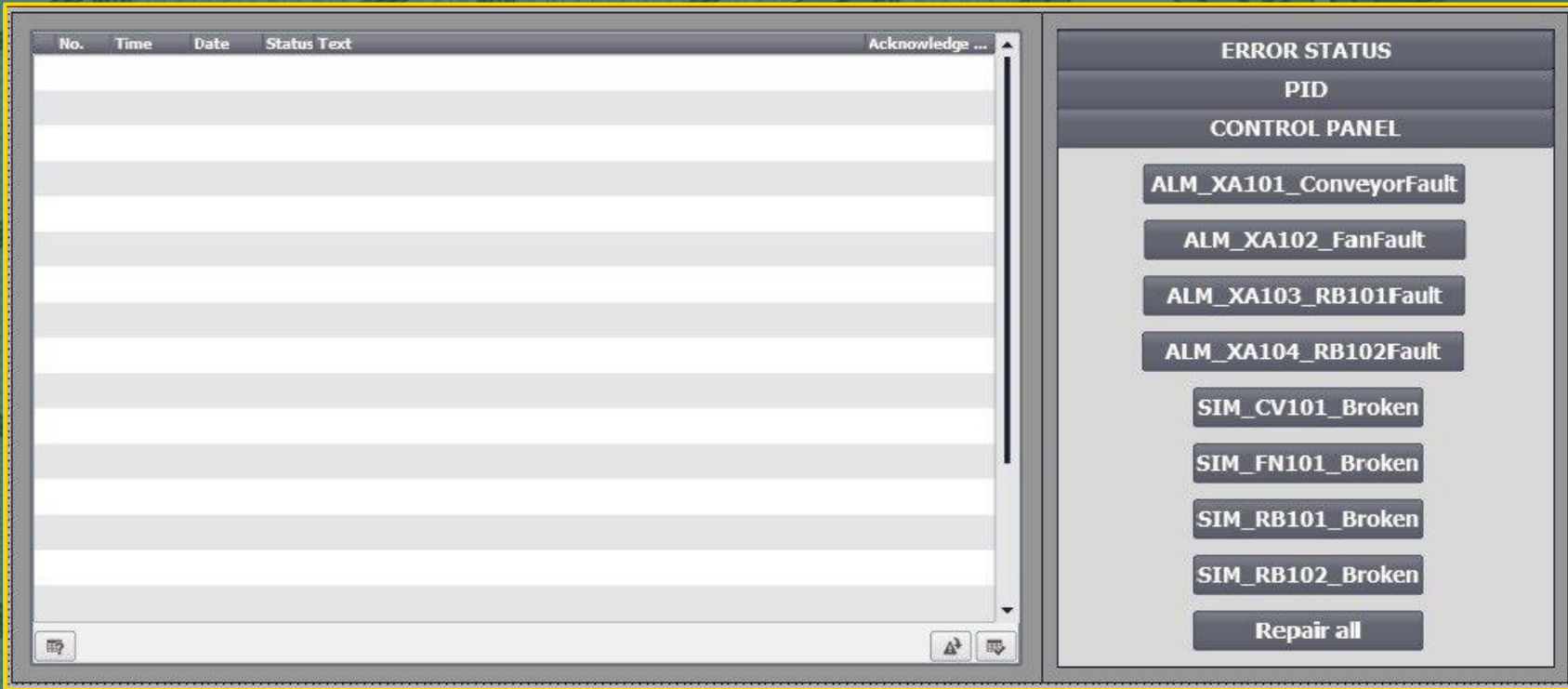


Figure 24 - HMI control panel for mode selection and system commands

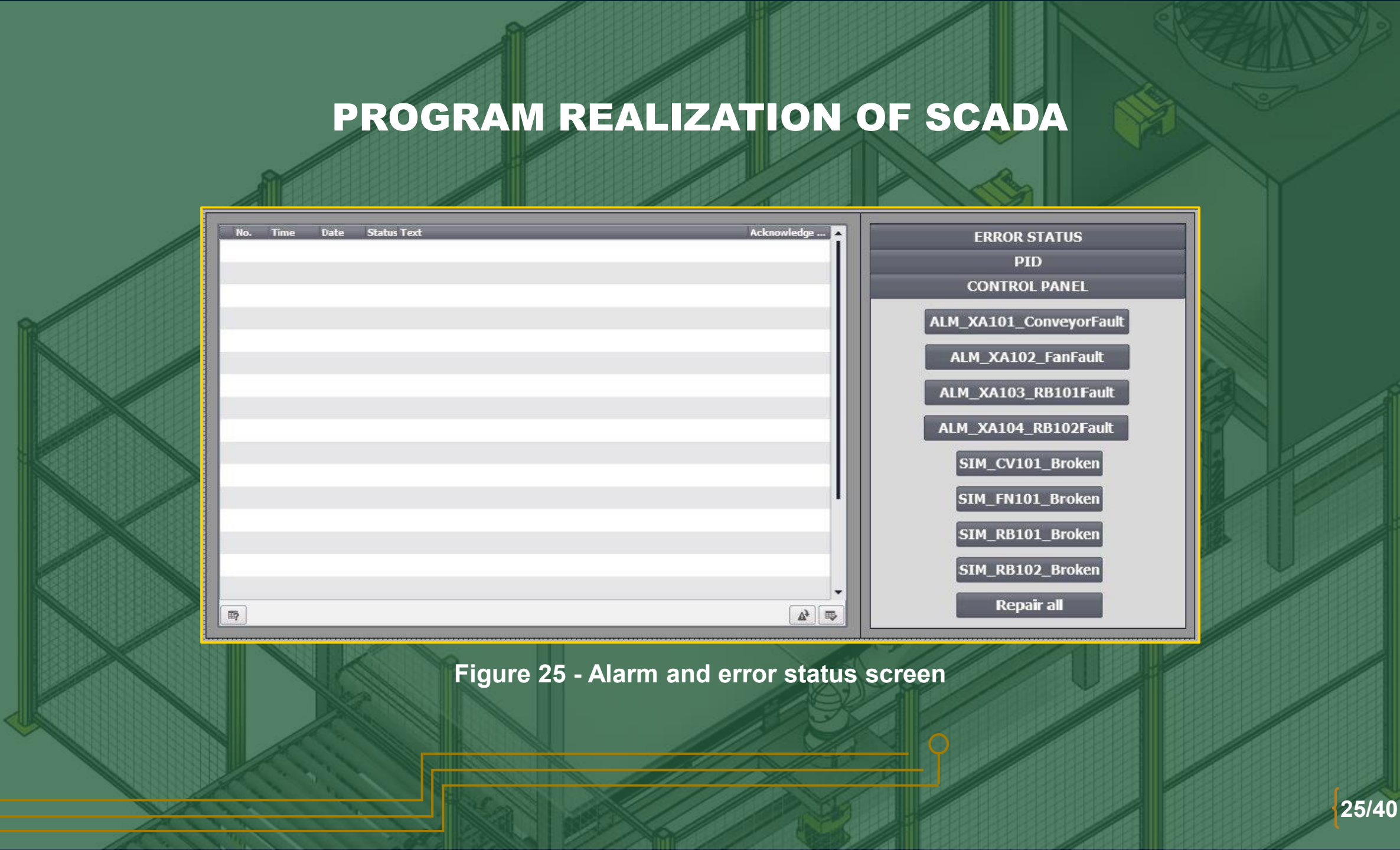
PROGRAM REALIZATION OF SCADA



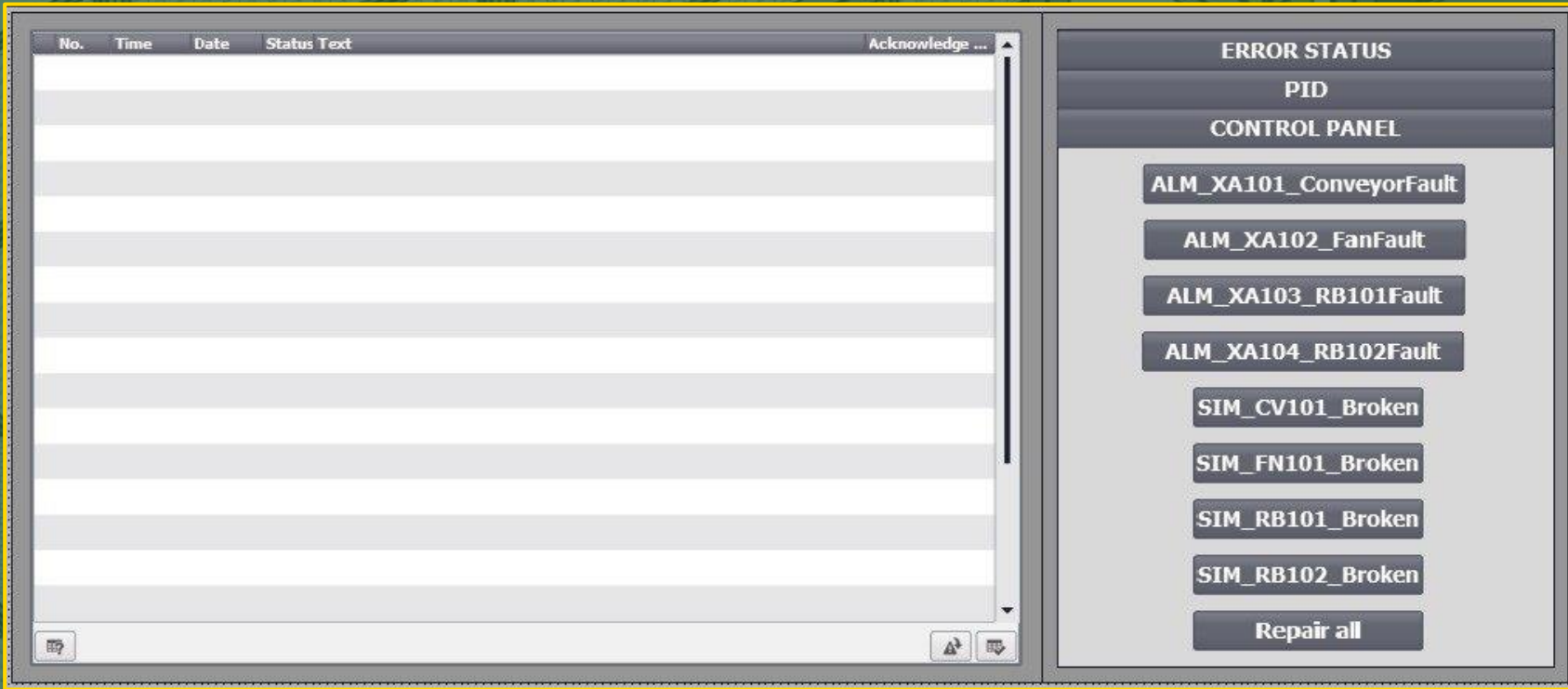
The screenshot displays the 'Alarm and error status screen' of a SCADA system. The interface is split into two main panels. The left panel features a table with the following headers: 'No.', 'Time', 'Date', 'Status Text', and 'Acknowledge ...'. The table body is currently empty. The right panel is titled 'ERROR STATUS' and contains a 'CONTROL PANEL' section. This section includes several buttons representing different system components and their status: 'ALM_XA101_ConveyorFault', 'ALM_XA102_FanFault', 'ALM_XA103_RB101Fault', 'ALM_XA104_RB102Fault', 'SIM_CV101_Broken', 'SIM_FN101_Broken', 'SIM_RB101_Broken', 'SIM_RB102_Broken', and a 'Repair all' button. The background of the slide shows a green industrial facility with conveyor belts and structural elements.

Figure 25 - Alarm and error status screen

25/40



PROGRAM REALIZATION OF SCADA



The screenshot displays the 'Alarm and error status screen' of a SCADA system. The interface is split into two main sections. The left section is a table with the following headers: 'No.', 'Time', 'Date', 'Status Text', and 'Acknowledge ...'. The table body is currently empty. The right section is titled 'ERROR STATUS' and contains a 'CONTROL PANEL' with several buttons: 'ALM_XA101_ConveyorFault', 'ALM_XA102_FanFault', 'ALM_XA103_RB101Fault', 'ALM_XA104_RB102Fault', 'SIM_CV101_Broken', 'SIM_FN101_Broken', 'SIM_RB101_Broken', 'SIM_RB102_Broken', and a 'Repair all' button. The background of the slide shows a green industrial facility with conveyor belts and structural elements.

Figure 25 - Alarm and error status screen

25/40

PROGRAM REALIZATION OF SCADA

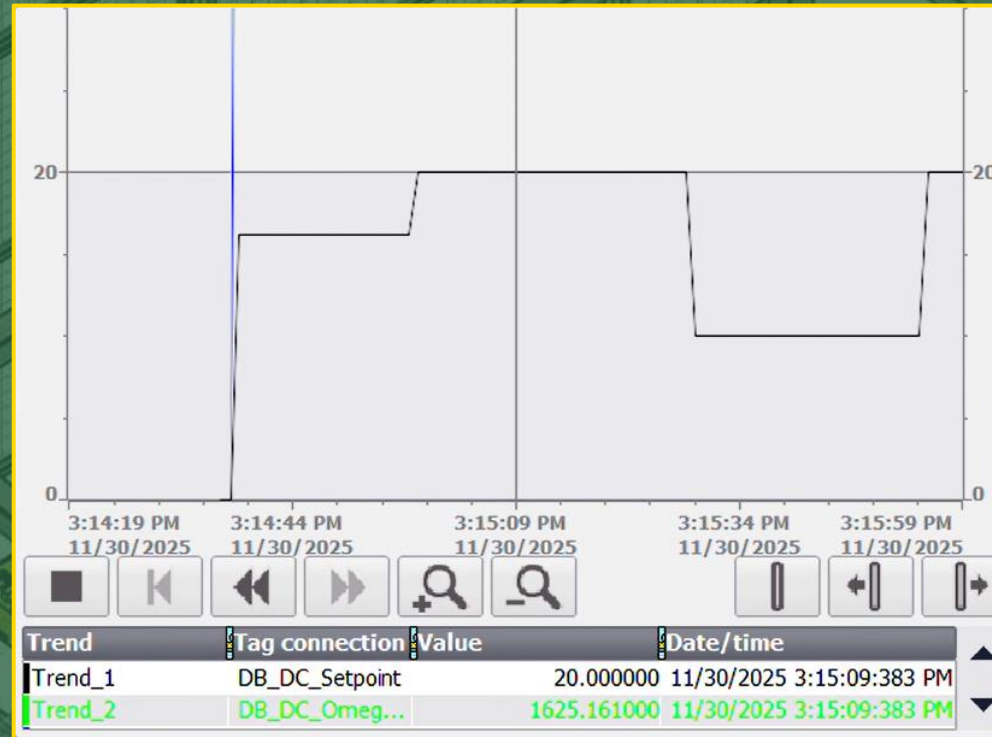


Figure 26 - SCADA interface for PID trend visualization

PROGRAM REALIZATION OF SCADA

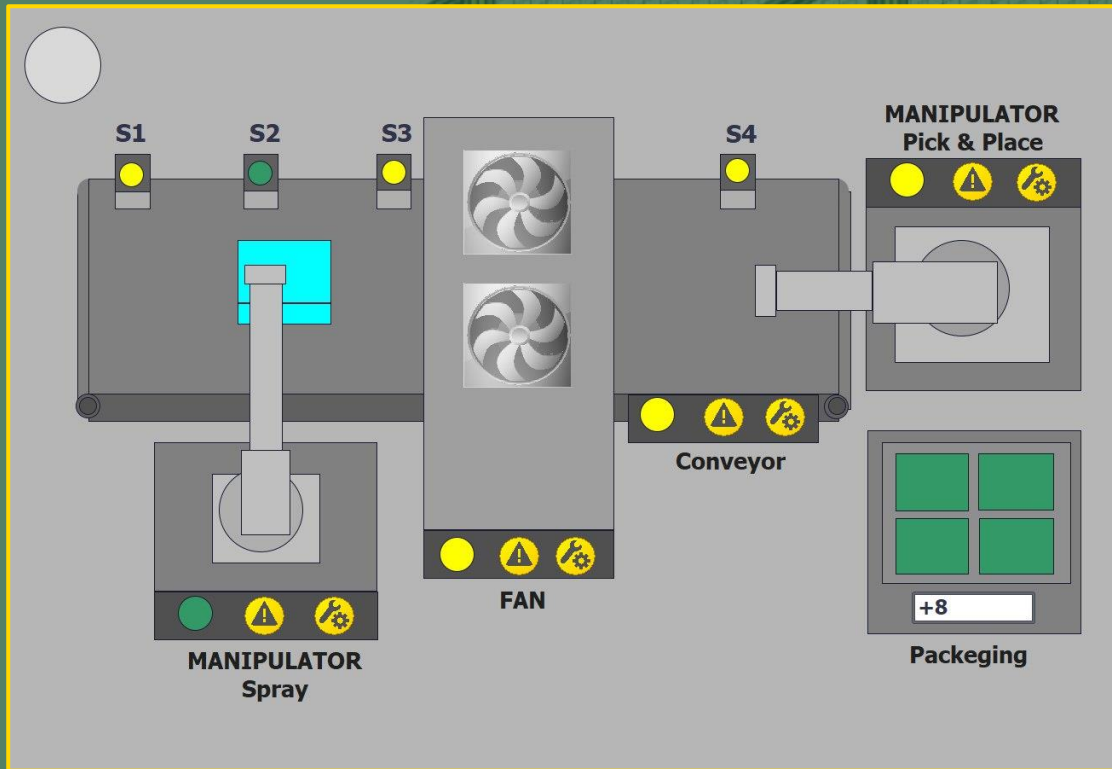


Figure 27 - HMI control panel for Automatic mode and Manual mode

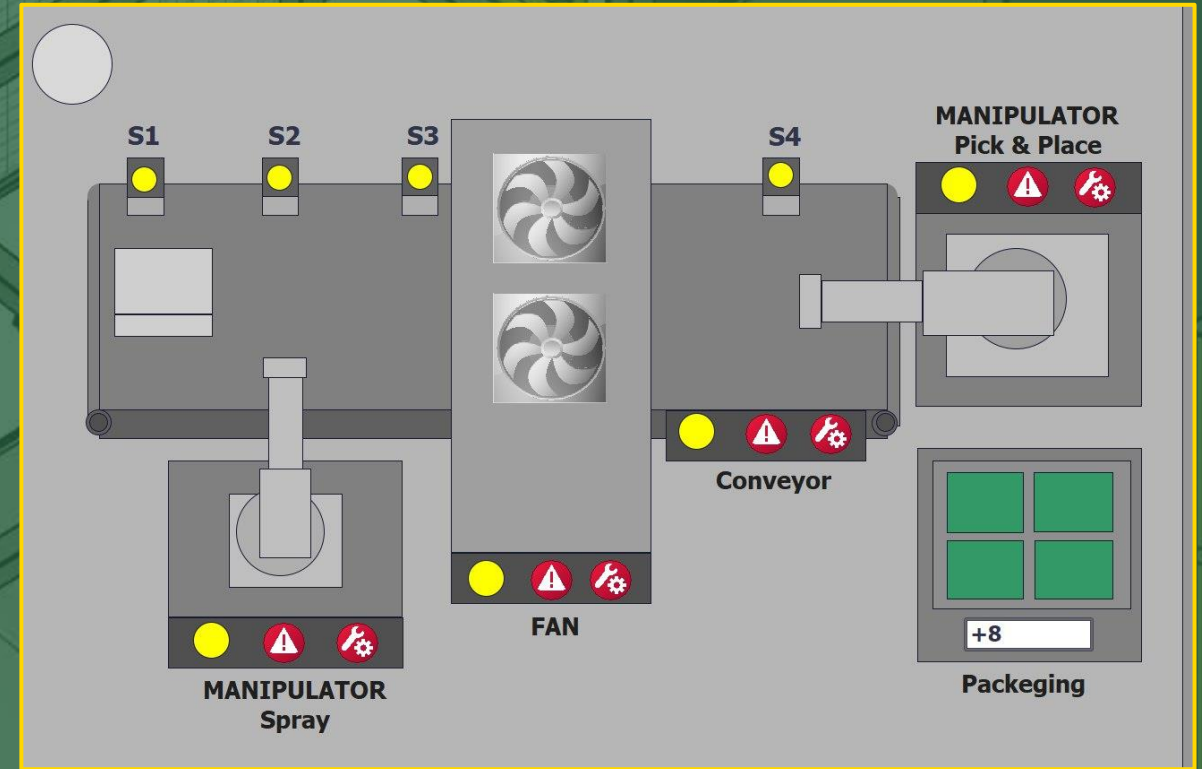
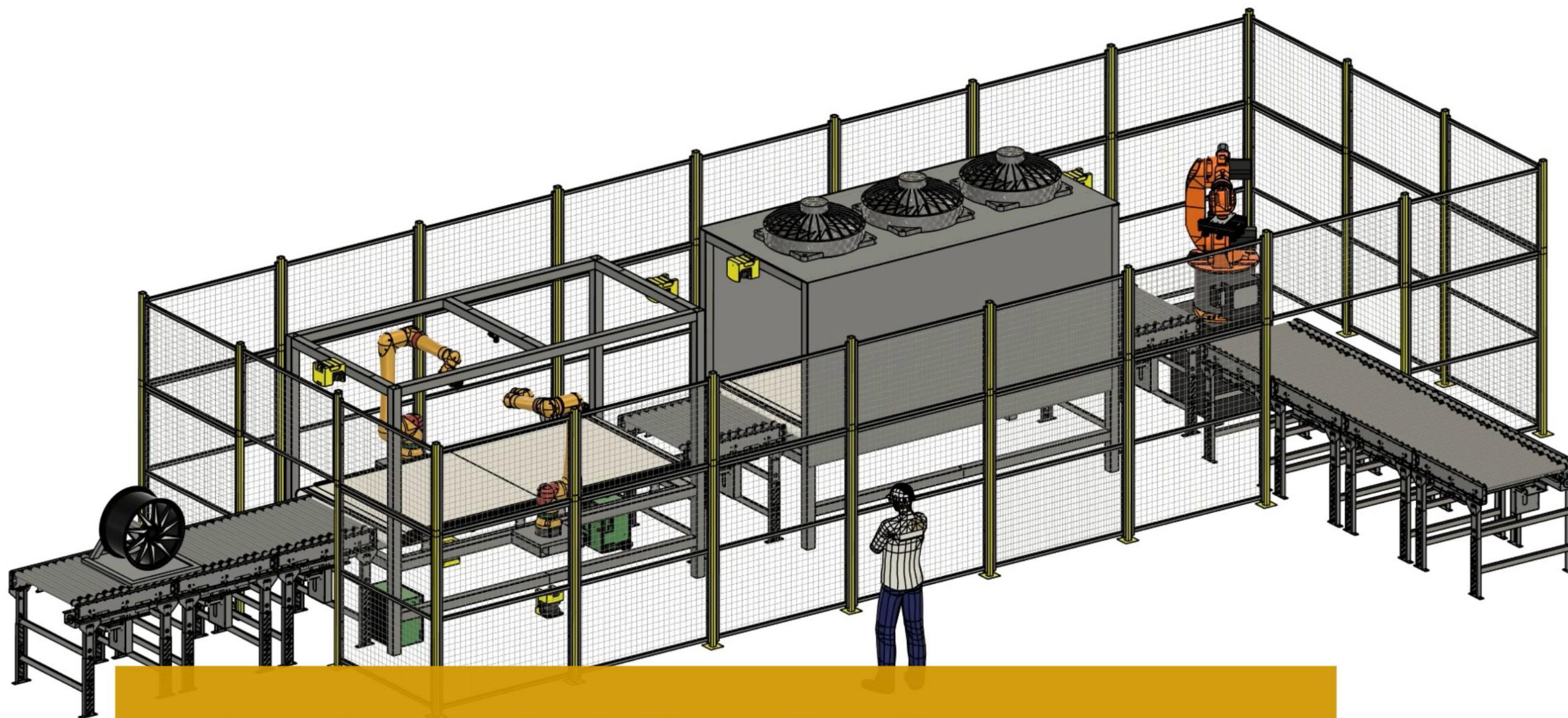


Figure 28 - HMI control panel for alarm and error status



Industrial Scale Process

DESIGNED DRAWINGS (AUTOCAD)

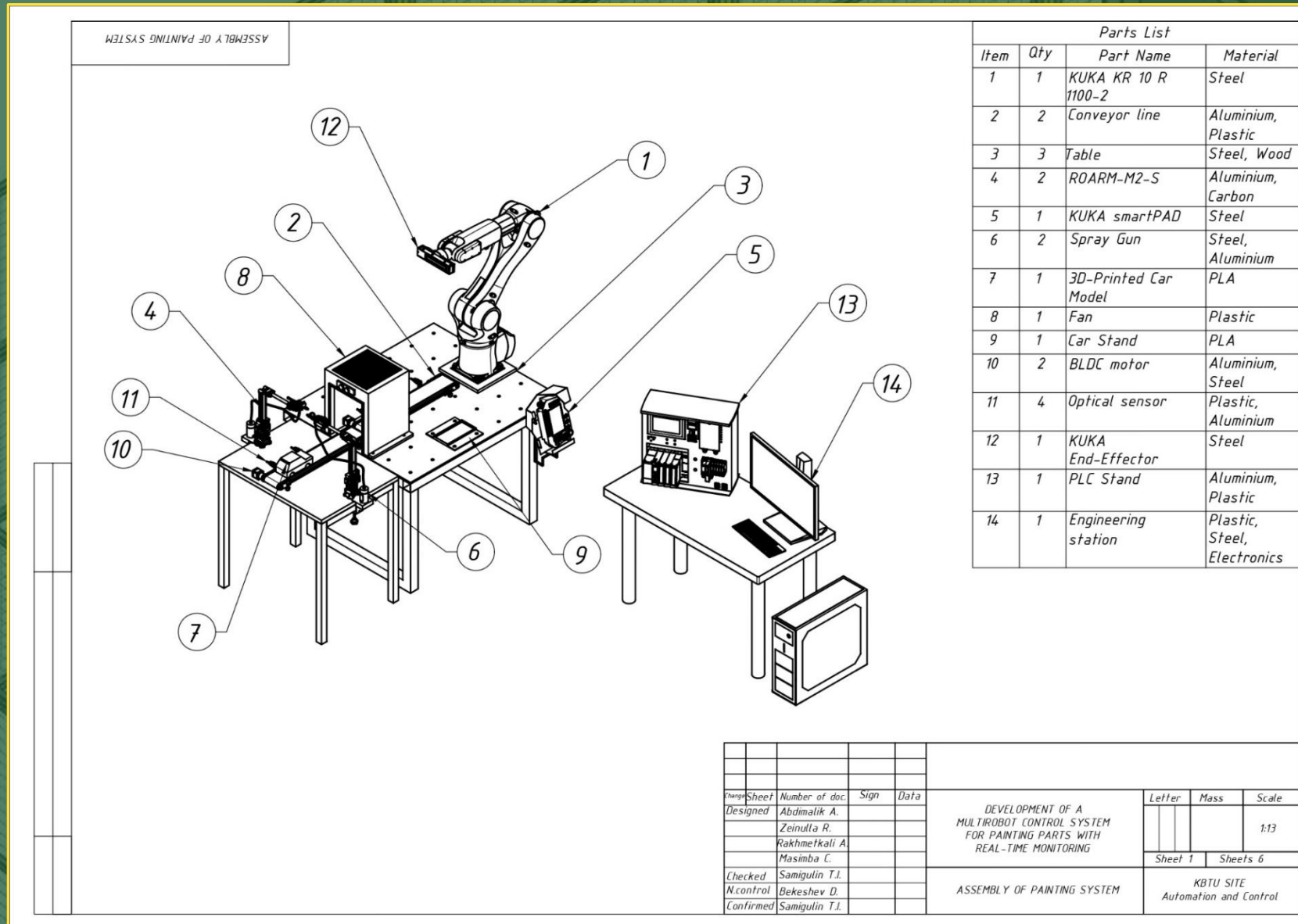


Figure 30 – Assemble of painting system

DESIGNED DRAWINGS (AUTOCAD)

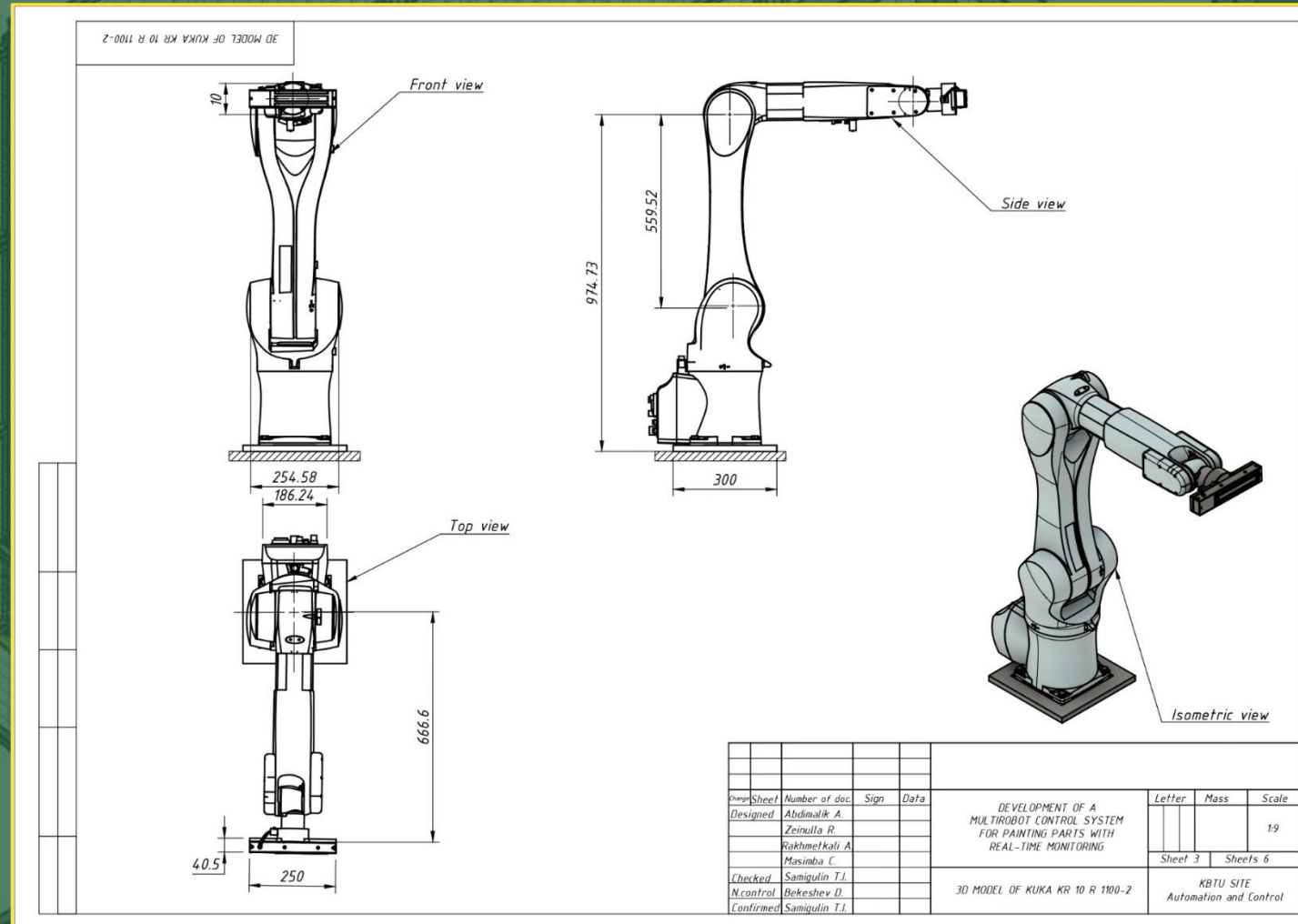


Figure 31 – 3D model of KUKA KR10 R1100-2

DESIGNED DRAWINGS (AUTOCAD)

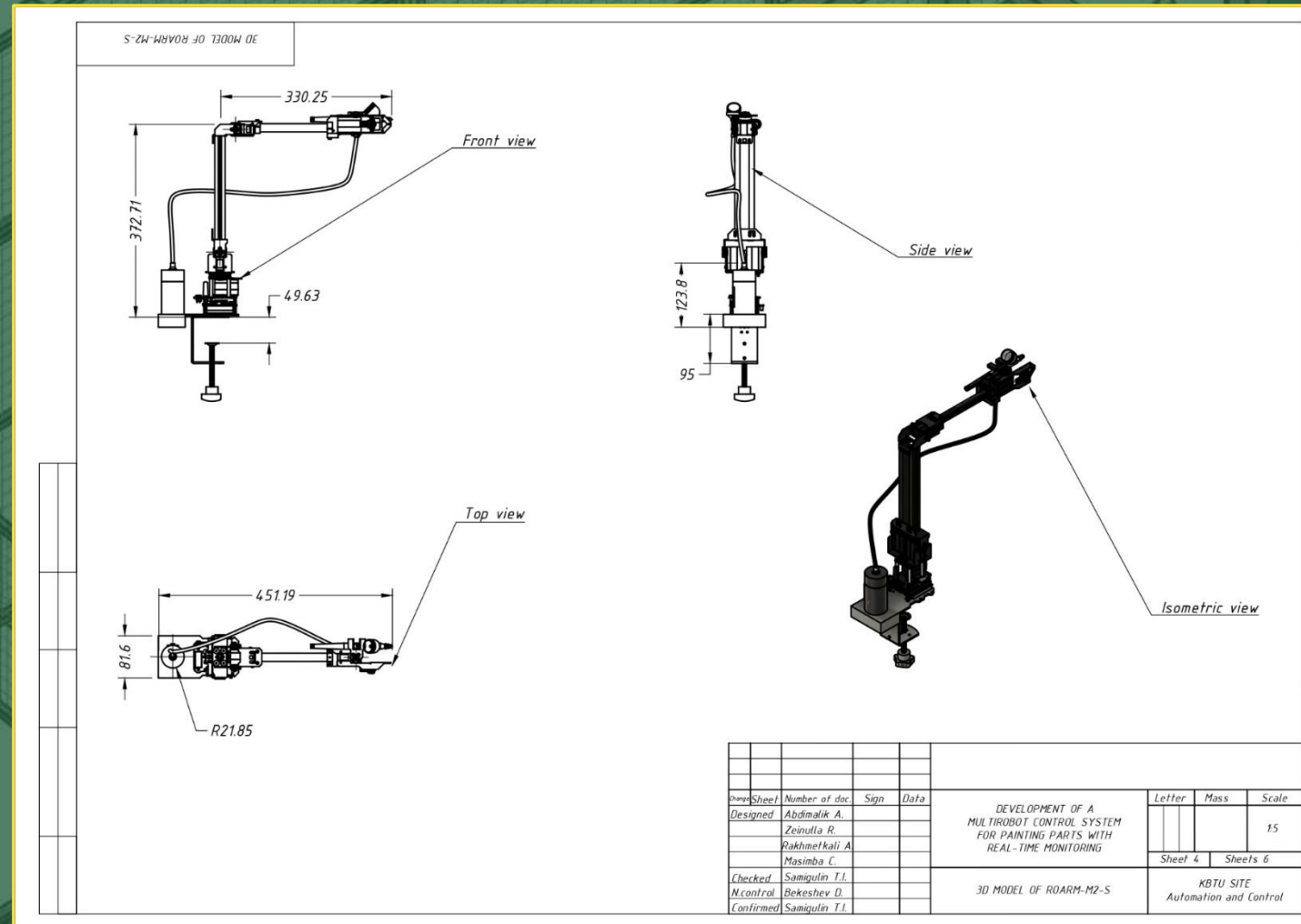


Figure 32 – 3D model of RoArm-M2-S

DESIGNED DRAWINGS (AUTOCAD)

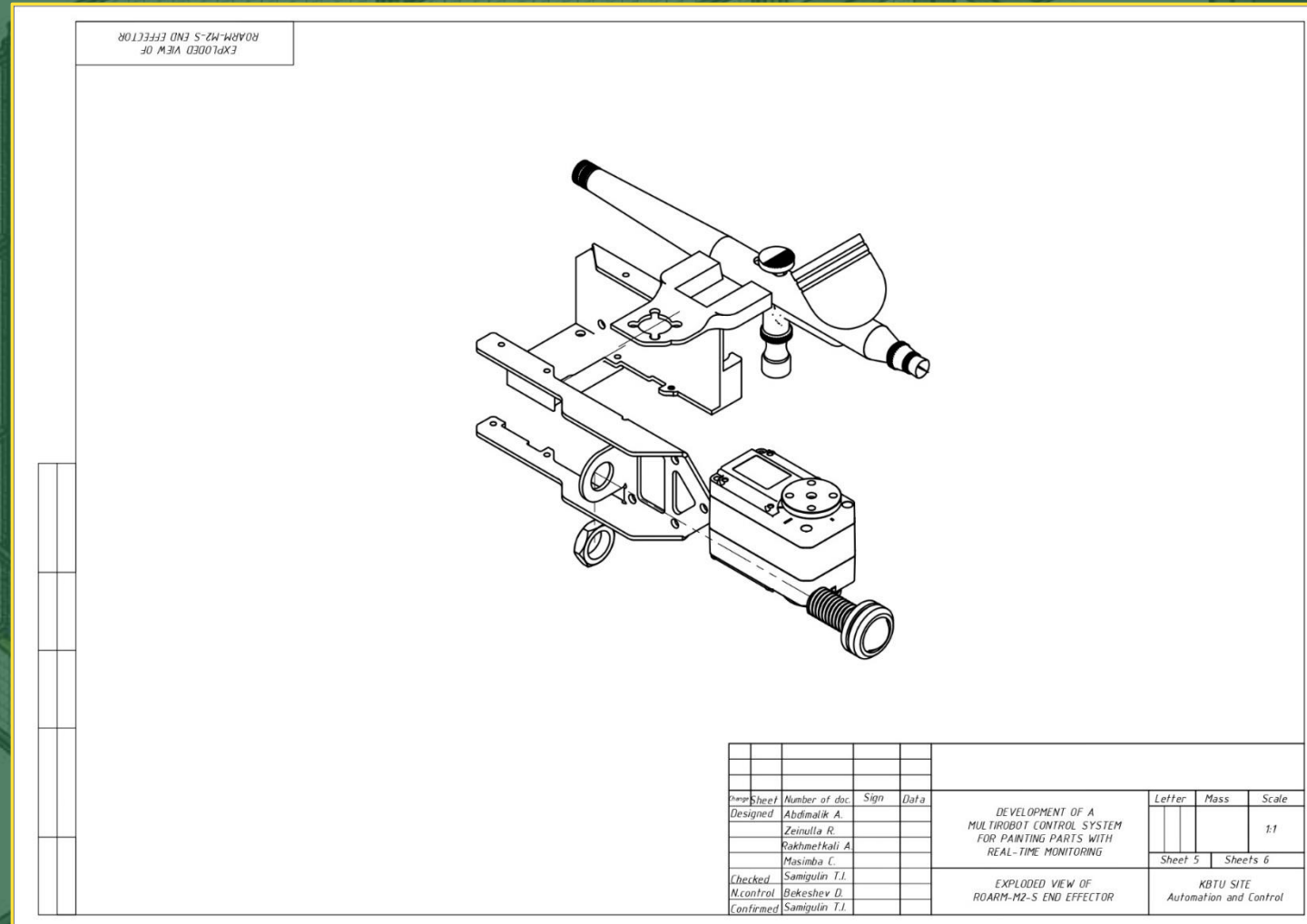


Figure 33 – Exploded view of RoArm-M2-S End Effector

DESIGNED DRAWINGS (AUTOCAD)

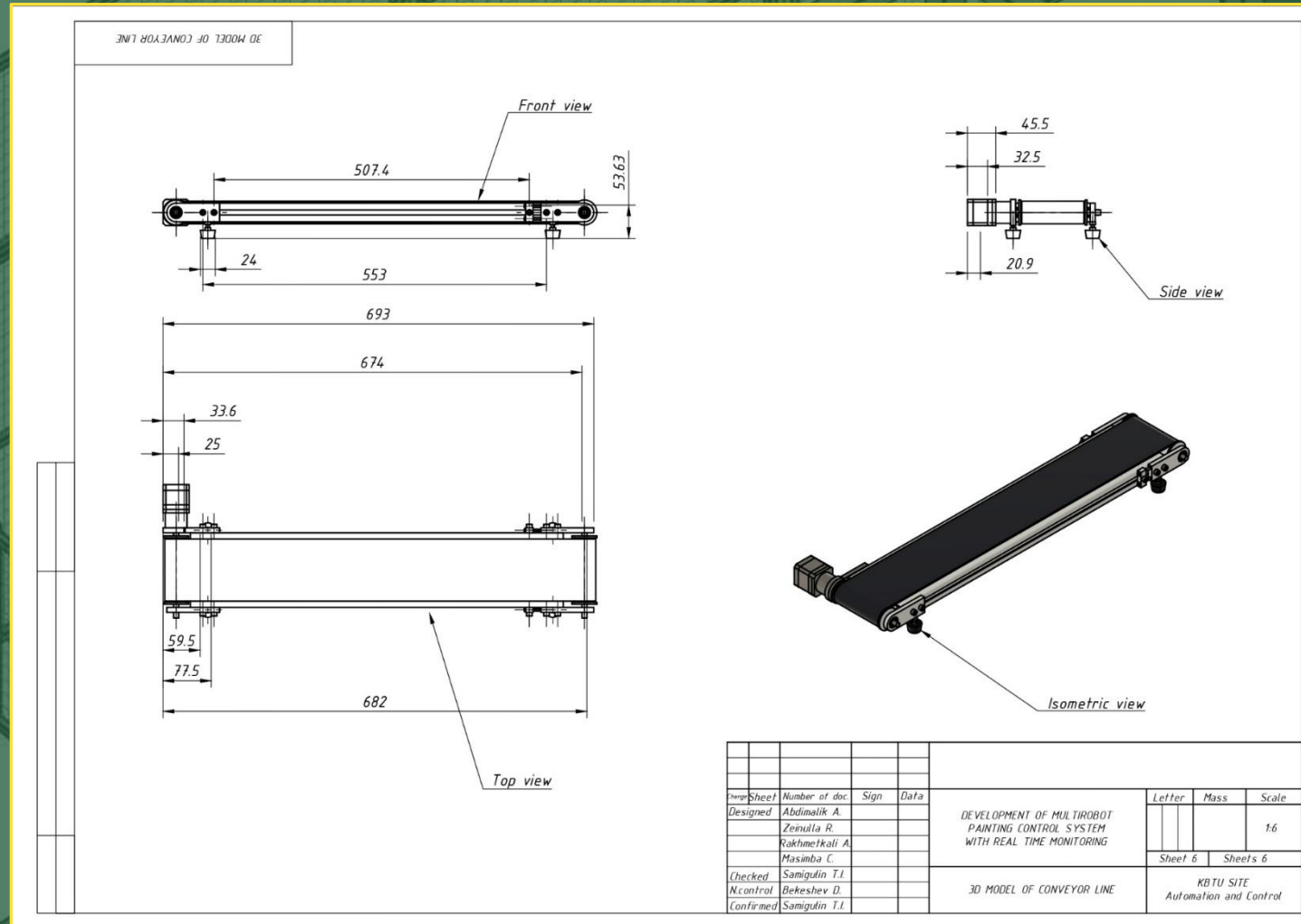


Figure 34 – 3D model of conveyor line

DESIGNED DRAWINGS (AUTOCAD)

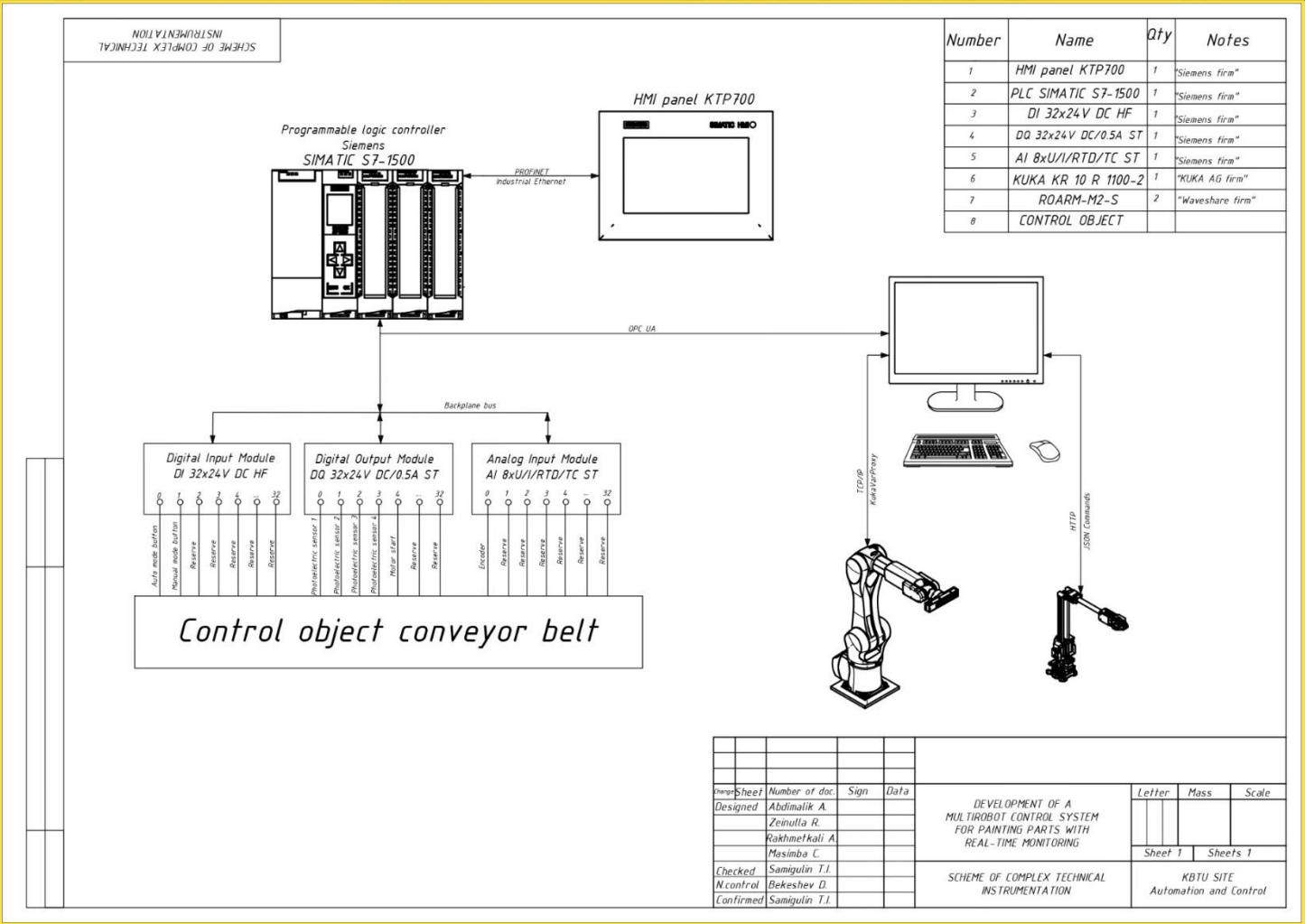


Figure 35 – Scheme of complex technical instrumentation

DESIGNED DRAWINGS (AUTOCAD)

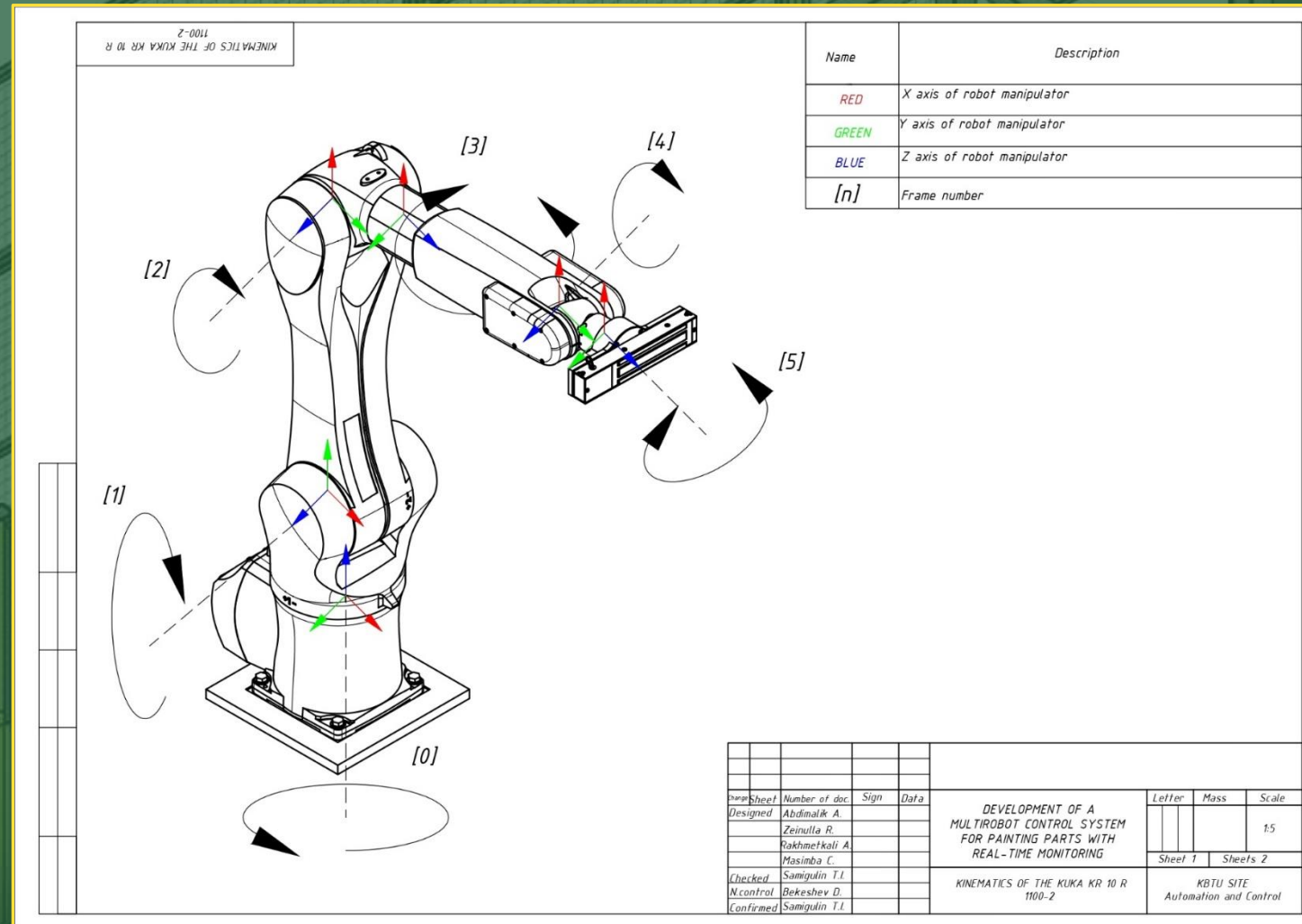


Figure 36 – Kinematics of the KUKA KR 10 R 1100-2/SEL

DESIGNED DRAWINGS (AUTOCAD)

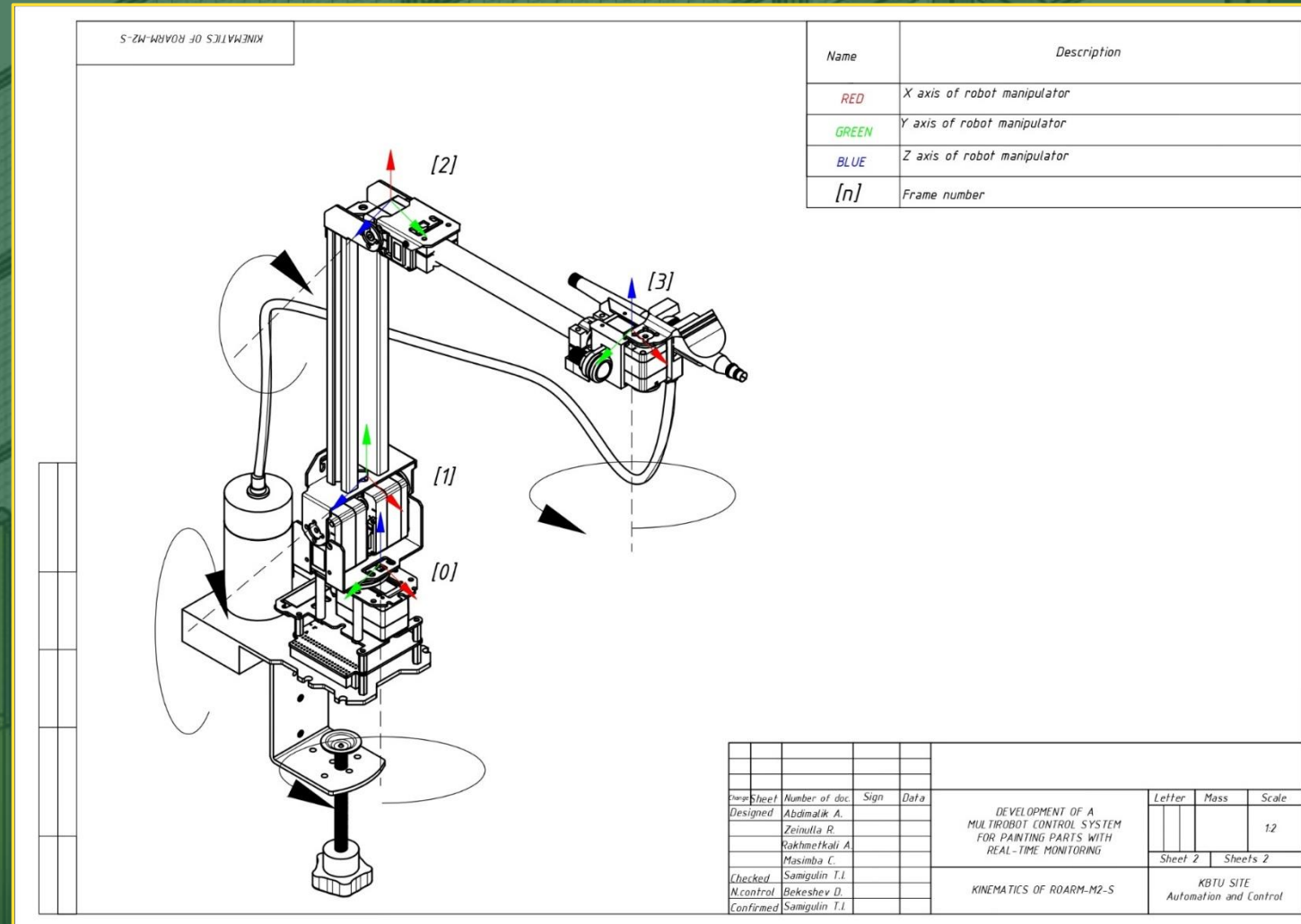


Figure 37 – Kinematics of the RoArm-M2-S

DESIGNED DRAWINGS (AUTOCAD)

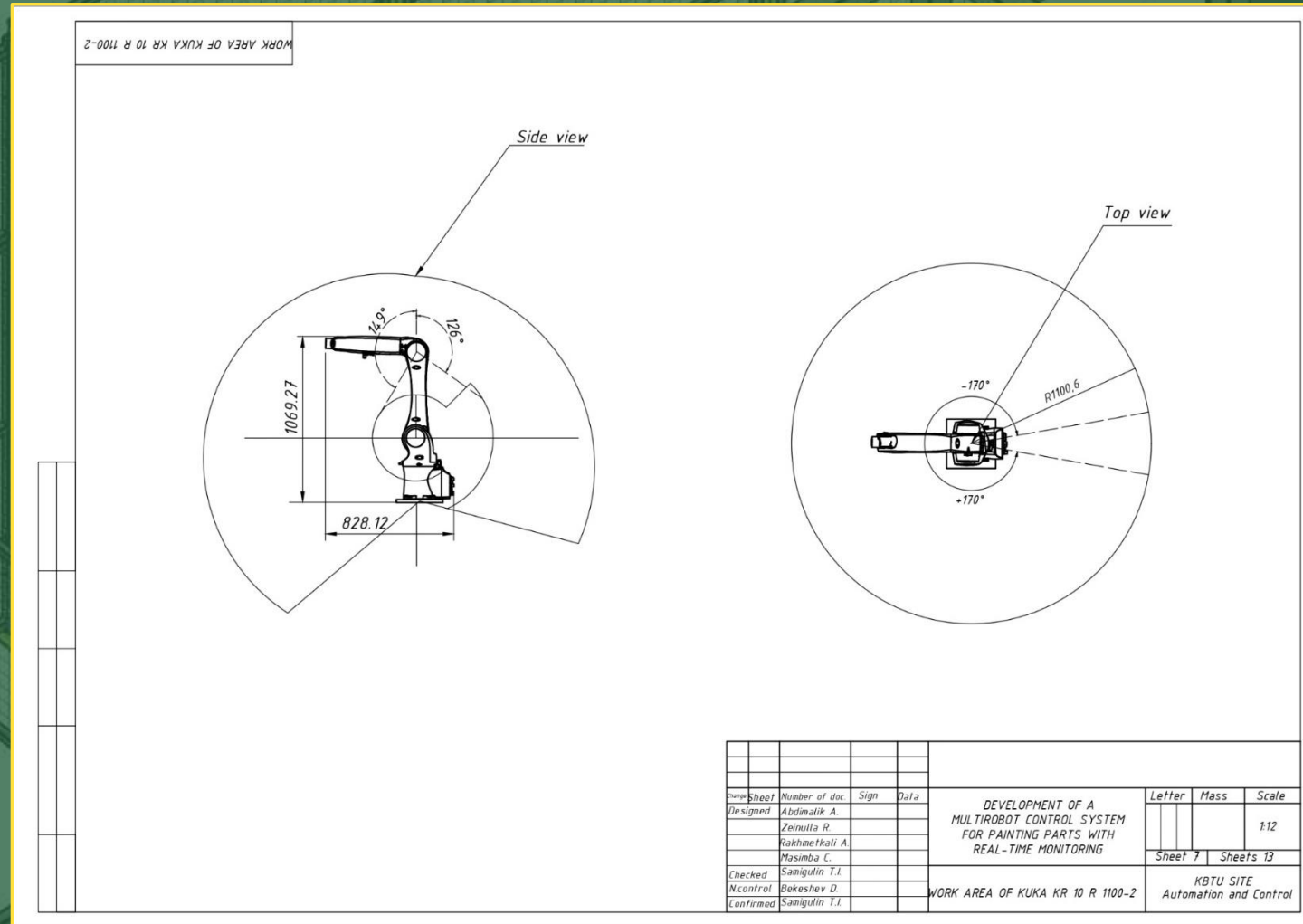


Figure 38 – Work area of KUKA KR 10 R 1100-2/SEL

DESIGNED DRAWINGS (AUTOCAD)

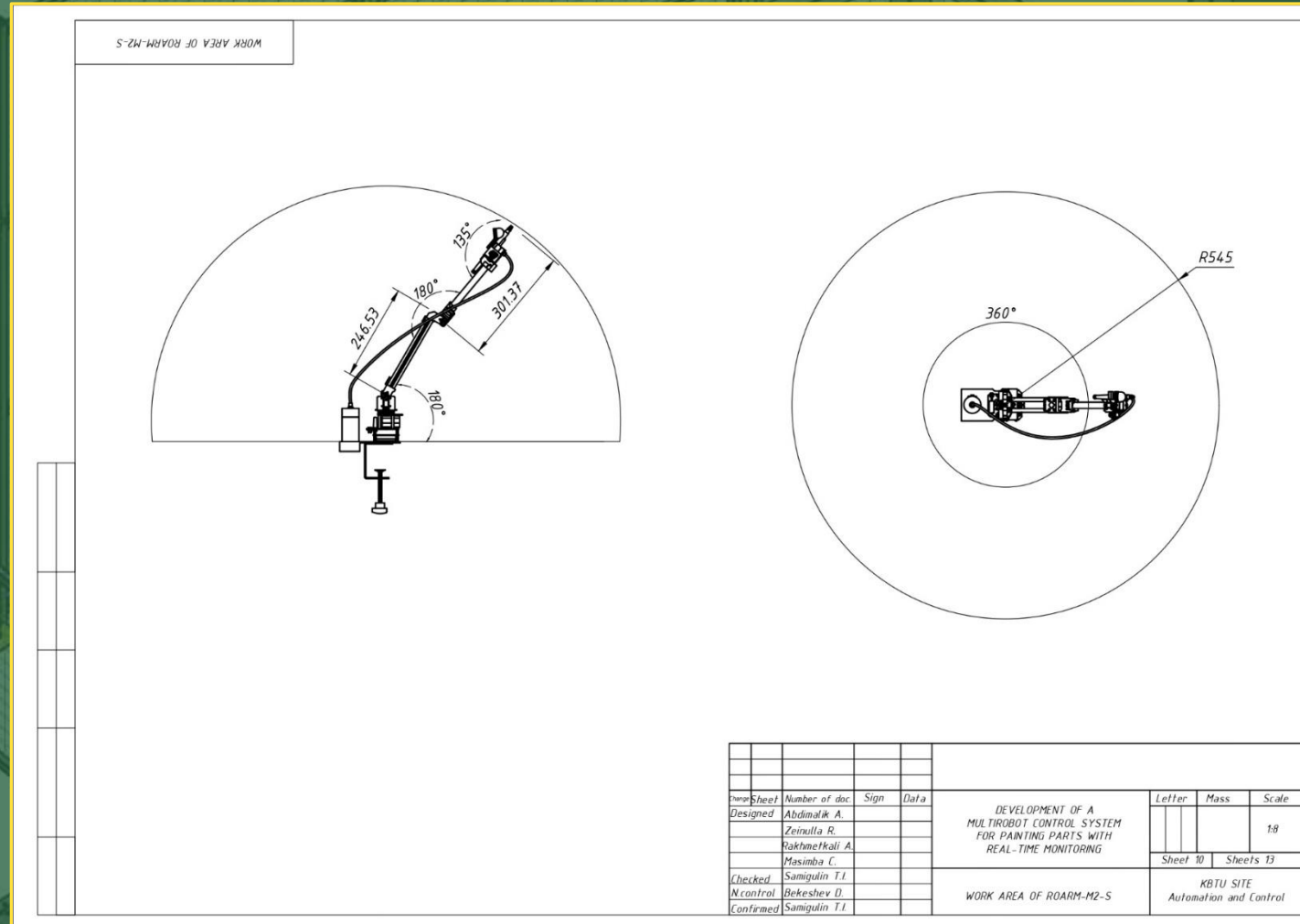


Figure 39 – Work area OF RoArm-M2-S

DESIGNED DRAWINGS (AUTOCAD)

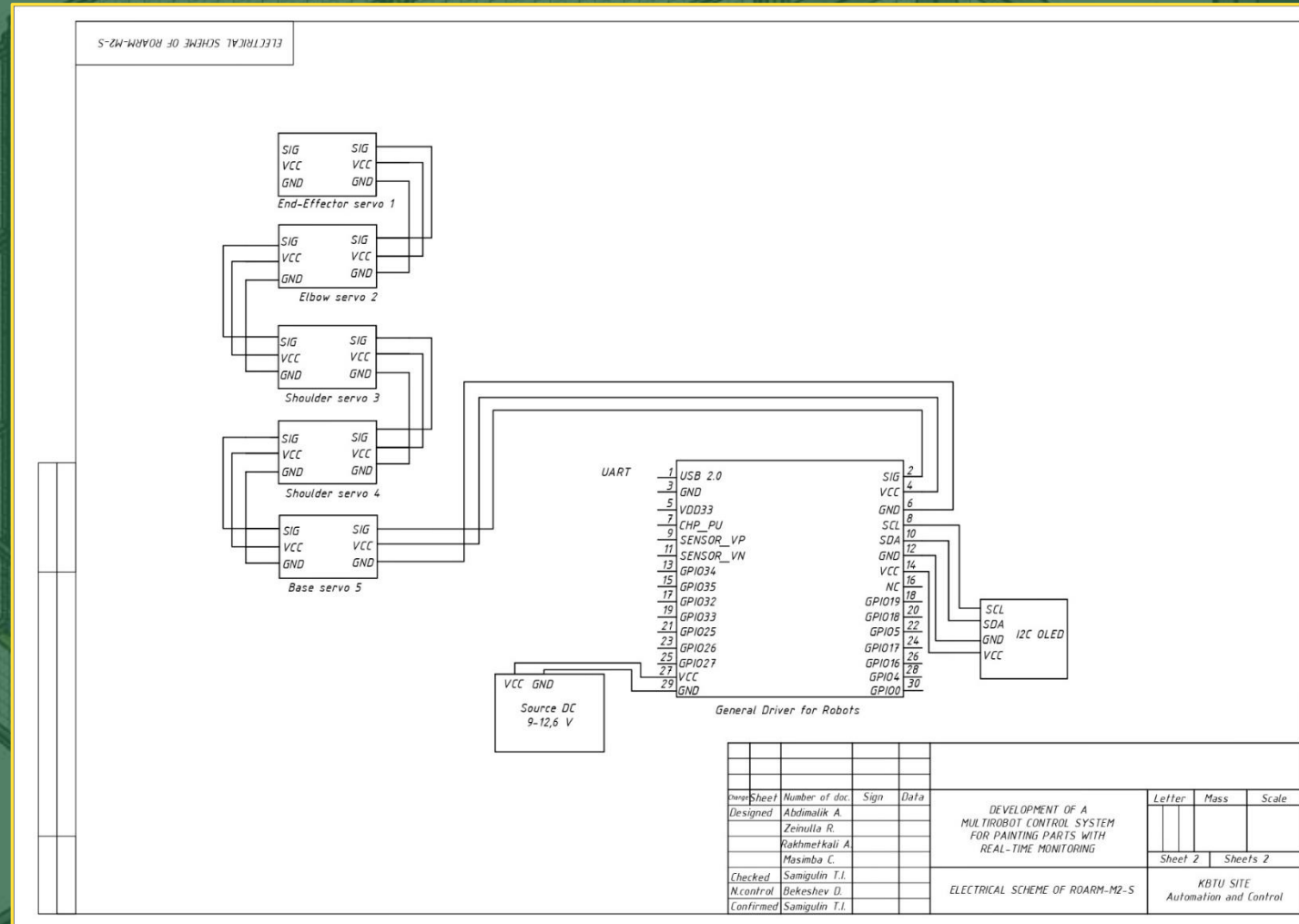


Figure 40 – Electrical Scheme of RoArm-M2-S

An isometric illustration of an industrial robotic cell, rendered in a dark green color scheme. The cell features a metal frame, safety fencing, and internal components like a robotic arm and various sensors. The text "THANK YOU FOR YOUR ATTENTION!" is centered in white. In the bottom right corner, there is a page number "40/40" enclosed in a bracket, with a yellow line graphic extending from the bottom left towards it.

**THANK YOU FOR YOUR
ATTENTION!**